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Improving forecasts for heterogeneous time series by "averaging", with application to food demand forecasts

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ABSTRACT

A common forecasting setting in real-world applications considers a set of possibly heterogeneous time series of the same domain. Due to the different properties of each time series, such as length, obtaining forecasts for each individual time series in a straightforward way is challenging. This paper proposes a general framework utilizing a similarity measure in dynamic time warping to find similar time series to build neighborhoods in a *k*-nearest neighbor fashion and improve forecasts of possibly simple models by averaging. Several ways of performing the averaging are suggested, and theoretical arguments underline the usefulness of averaging for forecasting. Additionally, diagnostic tools are proposed for a deep understanding of the procedure.

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1. Introduction

In many forecasting settings, one encounters a heterogeneous set of time series for which individual forecasts are required. A heterogeneous set of time series may imply time series of different seasonalities or trends. Hence, it may be difficult to model all of them in a joint way, or to choose an approach yielding reasonable results for the entire set of time series. Modeling each time series on its own might be an alternative way to handle this challenge, but this does not make use of the shared properties of the domain. Depending on this domain, each time series might be very difficult to model and forecast. Additionally, a practitioner may even want an automatic procedure to produce forecasts.

To this end, some work has been done on comparing local and global forecasting approaches. Whereas 'local' means modeling each time series on its own, 'global' models refer to modeling the set of time series simultaneously. Montero-Manso and Hyndman (2021) argue that global and local models can achieve the same forecasts without needing additional assumptions. The global approach does, however, require a rather high number of total observations, as well as additional hyperparameter tuning, and might be very difficult to find in the first place. In practice, there may be insufficient observations. On the other hand, local models are especially hard to use for short time series. Thus, we want to fill the gap of forecasting a set of heterogeneous time series that also contain short time series with a rather low number of total observations.

Hewamalage et al. (2022) investigate the notion of global models and the relatedness of the time series. They focus on simulation experiments controlling the data-generating process of each time series to allow for arbitrary relatedness in the set of time series as well. They then apply various machine learning methods to conclude that the performance and complexity of global models heavily depends on the heterogeneity of the data as well as the actual amount of available data. A set of very heterogeneous time series requires possibly very complex global models, and this in turn requires a lot of overall data. This leads us again to the challenge mentioned above.

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The shortcomings of a totally global model are addressed by Godahewa et al. (2021). The authors argue that with an increasing amount of data, a global model may be insufficiently localized, leading to worse forecasts. Thus, they propose a localization technique where the time series are clustered and each cluster is then individually modeled by a global model. The clusters are feature- or distance-based, or even randomly assigned. While sounding similar to this paper's work, the approach differs from ours. We do not model each neighborhood of time series. Rather, we use the neighbors' existing models. Similarly, Bandara et al. (2020) discuss the use of neural networks on feature-based time series clusters, but these require a lot of data. Another approach combining local and global modeling is taken by Smyl (2020), who combines neural networks and statistical models such that the neural network is modeled across all time series, and local behavior is modeled by exponential smoothing models. This approach won the M4 forecast competition (Makridakis et al., 2020).

In terms of short time series, there is little literature available to tackle this challenge. Cruz-Nájera et al. (2022) compare simple and machine learning-based models on a set of short time series (14 to 21 observations each) regarding crimes in Mexico. The authors conclude that simple models like the simple moving average or ARIMA perform better than more complex models such as neural networks.

Considering all mentioned aspects, our contributions are as follows. This paper proposes a meta-framework for forecasting a set of possibly very heterogeneous time series by utilizing a range of local models and aggregating them in an appropriate way, exploiting the similarities of those time series. The similarities can be seen as homogeneous parts of the time series. Hence, we require some homogeneity in the heterogeneous set of time series.

In fact, every time series in a dataset is modeled by a local model. For each time series, we obtain a set of forecasts using the estimated models of its neighboring time series, and perform a type of model averaging to improve its forecasts. This differs from regular model averaging where one time series is modeled by a variety of models. This approach is beneficial in that we firstly allow for simple models in case of short time series, and still do not require a large total number of observations, as in the case of complex global models. Therefore, we can say that our methodology is a mixture approach: using local models while still taking into account the whole set of time series in the forecast procedure. This procedure remains very flexible, since we do not fix any family of possible models. Even the use of more complex models such as neural networks is still possible.

The rest of the paper is structured as follows. In Section 2.1 we introduce the measure of (dis-)similarity as it is used in our methodology, namely dynamic time warping (DTW). More details are available in Appendix A.1. Section 2.2 introduces the notion of k-nearest neighbors in the context of time series, while in Section 2.3 we propose several averaging methods to improve forecasts. In Appendix A.2 we shortly outline how DTW can be used to obtain an average time series. A simple theoretical motivation is given in Appendix A.3, where we validate the notion of model averaging under certain assumptions. Section 3 is about the evaluation techniques we apply, followed by experiments in Section 4, where five different datasets are modeled and deepening diagnostics are performed for one of the datasets. Concluding remarks are given in Section 5.

2. Methodology

In this paper, we propose the following methodology. Given a finite set of time series $\mathcal{T}$ of possibly different properties, we first propose an appropriate dissimilarity measure on $\mathcal{T}$ by using asymmetric open-begin open-end dynamic time warping; see Section 2.1. This dissimilarity measure is used to construct neighborhoods for each time series in $\mathcal{T}$.

Next, for a fixed time series, its neighbors are aggregated in a new way to obtain improved forecasts. This aggregation is proposed to be done in multiple ways based on k-nearest neighbors (Section 2.2). The actual model averaging is described in Section 2.3. Since it might not be clear how DTW corresponds to actual forecasts, we give a theoretical motivation in Appendix A.3 where the relationship between DTW and forecast distributions is studied for simple state-space models.

A general overview of the methodology given in the form of a flowchart is seen in Fig. 1.

2.1. Dynamic time warping

A common technique for measuring the similarity of two time series is dynamic time warping (DTW), introduced by Sakoe and Chiba (1978). Originating from speech recognition, the goal is to align two sequences using a time-warping function in a cost-minimizing manner. In detail, we consider the (possibly multivariate) sequences $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n)' \in \mathbb{R}^{n\times d}$ and $\mathbf{Y} = (\mathbf{Y}_1, \dots, \mathbf{Y}_m)' \in \mathbb{R}^{m\times d}$ with possibly $n \neq m$ with appropriate dissimilarity function on the rows of $\mathbf{X}$ and $\mathbf{Y}$ as $d(i,j) := d(\mathbf{X}_i, \mathbf{Y}_j)$. Denote the entries of $\mathbf{X}_i$ and $\mathbf{Y}_j$ by X_{ik} and Y_{jk}, respectively, where $k = 1, \dots, d$. A common choice, also used in this paper, is the Euclidean distance, i.e. $d(i,j) = \sqrt{\sum_{k=1}^{d}(X_{ik} - Y_{jk})^2}$. The DTW distance is now defined as

$$\mathrm{DTW}(\mathbf{X}, \mathbf{Y}) := \min_{\phi\in\Phi} \sum_{k=1}^{K_\phi} w_\phi(k) d(\phi_{\mathbf{X}}(k), \phi_{\mathbf{Y}}(k)), \tag{1}$$

where $\phi = (\phi_{\mathbf{X}}, \phi_{\mathbf{Y}}) : \{1, \dots, K_\phi\} \to \{1, \dots, n\} \times \{1, \dots, m\}$ denotes the warping function, and w_ϕ denotes the corresponding weights. The warping function ϕ links the two sequences in a cost-minimizing way. For example, $\phi(k) = (\phi_{\mathbf{X}}(k), \phi_{\mathbf{Y}}(k))$ implies that $\mathbf{X}_{\phi_{\mathbf{X}}(k)}$ and $\mathbf{Y}_{\phi_{\mathbf{Y}}(k)}$ are linked together. The length of the warping function K_ϕ depends on the optimal warping function and is therefore chosen alongside the minimization problem in (1).

A more in-depth introduction to DTW is given in Appendix A.1, containing details about the warping functions as well as computing a representative time series using DTW.

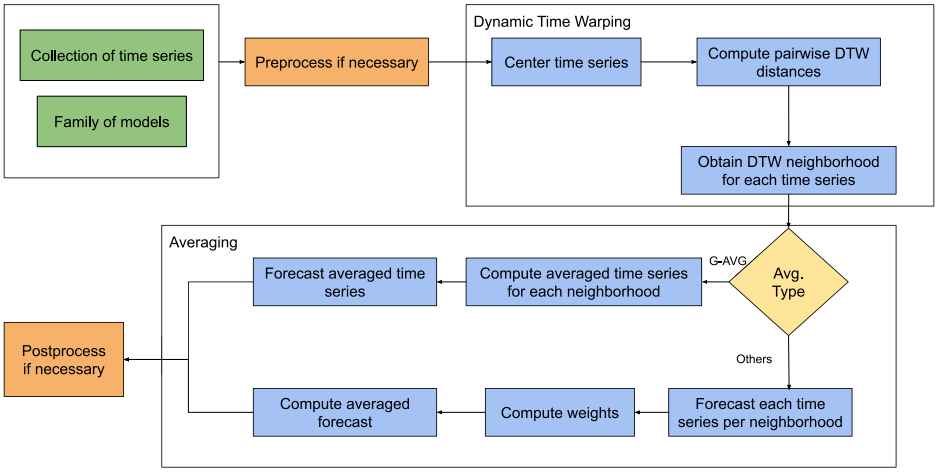


Fig. 1. Overall workflow of the proposed averaging methodology.

2.2. Nearest neighbors

For many classification and regression problems, k-nearest neighbors (k-NN) is an easy yet well-performing method to apply and improve predictions. The basic idea is as follows. Consider a metric space (X, δ) and $x \in X$. Then, a neighborhood around x can be formed by computing $\delta(x, z)$ for each $z \in X, z \neq x$. The k elements with minimal distance to z are then the neighbors of x. Eq. (2) gives a formal definition for this.

$$\mathcal{N}(x) := \{z_1, \dots, z_k : \delta(x, z_1) \leq \cdots \leq \delta(x, z_k) \leq \delta(x, z), \; z \neq z_i, i = 1, \dots, k\}. \tag{2}$$

In case of ties, one could increase the neighborhood's size or choose one of the equally distanced neighbors randomly. For the neighborhood including x itself, we write $\overline{\mathcal{N}}(x) := \mathcal{N}(x) \cup \{x\}$.

From a statistical point of view, we consider observations $(x_1, y_1), \dots, (x_N, y_N) \in X \times Y$, where Y denotes the set of possible labels and X the feature space. We write $y(x)$ if y is the label corresponding to x. In an unsupervised framework there would not be any labels and we would just be able to form neighborhoods. In a classification setting, Y is a discrete set, whereas the easiest example is a binary classifier with $Y = \{0, 1\}$. A new observation x_0 may be classified as the majority class of its neighborhood. In a regression setting, the labels are continuous, e.g. $Y = \mathbb{R}$, and the predicted label is set to be $\hat{y}(x_0) = f(\overline{\mathcal{N}}(x_0))$, where f is an aggregation function. In the simplest case we have $\hat{y}(x_0) = \frac{1}{k} \sum_{x \in \mathcal{N}(x_0)} y(x)$, where the predicted label is just the arithmetic mean of the neighborhood's labels.

The choice of k is vital, but it is still not completely clear how to choose it. There are many heuristics, such as choosing $k \approx \sqrt{N}$, which do not seem reasonable in many applications. In contrast, the optimal k can be chosen based on an optimization criterion. In supervised settings, one usually splits the data into training and test sets, and chooses k to minimize a loss function on the training set. This k is then fixed and used to predict on the test set. For more advanced approaches where k is determined adaptively, see Anava and Levy (2017).

2.2.1. Time series nearest neighbors

In the context of time series, k-NN has also been widely used. In time series classification, 1-NN is often used in conjunction with DTW (Jeong et al., 2011). Many approaches also consider k-NN in a feature space. Such a feature space could consist of time series features like the length, trend, auto-correlation properties, and many more. For such cases, DTW is not even used. However, this approach assumes that the features can be extracted in a reasonable way, which might not always be the case. In terms of k-NN for regression or forecasting, Martínez et al. (2017) use a simple approach where a single time series can be forecasted by the mean value of neighboring labels based on Euclidean distances of lagged values. Naturally, this method can also be applied to many time series in a pooled manner, but this would not use the idea of employing similar time series for improving forecasts, where similarity is based on DTW.

2.2.2. Our setting

We let X be a set of heterogeneous and differently sized time series, equipped with the asymmetric open-begin open-end DTW distance measure. Note that this space is not metric. However, k-NN can still be applied in such a setting. The label set Y is not uniquely defined. For 'first model, then average' models (see Section 2.3), this set consists of one-step-ahead forecasts for the time series of interest. In the case of the 'first average, then model' approach, the label set and the aggregation function are more complicated and are described in the corresponding section.

2.3. Model averaging

For the upcoming sections, let $\delta(x, y) = \mathrm{DTW}_{\text{asym, OBE}}(x, y)$. Further, let $\hat{\cdot} : (x, M) \mapsto \hat{x}_{t+1|t,\dots,s}(M)$, where $x = (x_s, \dots, x_t)$ is a time series and M is any model used to forecast x_{t+1}. That is, $\hat{x}_{t+1}(M) := \hat{x}_{t+1|t,\dots,s}(M)$ is the

one-step-ahead forecast of x obtained by using model M. We refer to M as the baseline model which we aim to improve.

We understand a model M as a mapping $M : \mathcal{T} \to \mathcal{M}$, taking a time series $x \in \mathcal{T}$ and outputting $M_x \in \mathcal{M}$, where $\mathcal{M}$ denotes the set of all possible models.

Given a model $M = M_x$ and a time series z, we denote $\tilde{\cdot} : (z, M_x) \mapsto \tilde{M}_x(z)$ as the refitting function. That is, $\tilde{M}_x(z)$ is the model M_x, estimated on x but fitted on z. More specifically, M_x is first trained on x, yielding estimated parameters. Fixing those estimated parameters, we now plug in z to obtain fits of the model on z. No data of z are used to estimate parameters. Thus, forecasts of $\tilde{M}_x(z)$ are always with respect to z.

2.3.1. 'First average, then model' approach

The first approach to improving a single forecast yielded by a baseline model is as follows. Given a time series $y = (y_s, \dots, y_t)$, we find a neighborhood $\mathcal{N}_y = \{z_1, \dots, z_{k_y}\}$ based on k-NN. Since the DTW distance also depends on the difference of the level in the time series, we center each time series first to avoid this effect. That way we keep other properties of the time series such as trends or seasonalities, since these might still be helpful to improve forecasts.

Denote by $z^c := (z_{s_z} - \bar{z}, \dots, z_{t_z} - \bar{z})$ the centered time series for $z = (z_{s_z}, \dots, z_{t_z})$ with $\bar{z} = \sum_{i=s_z}^{t_z} z_i/(t_z - s_z + 1)$. Then, a neighborhood around y is constructed such that $\delta(y^c, z_1^c) \le \delta(y_c, z_2^c) \le \cdots \le \delta(y, z_{k_y}^c)$ and $z_i = (z_{s_i}, \dots, z_{t_i})$ with $t \ge t_i$, $t_i - s_i > t - s$. That means we look for neighboring time series which have more historical data. This is especially motivated by possibly having shorter time series that will be difficult to model and is often the case in many online forecasting settings where data of a subset of the time series are rare. In such cases, it is not reasonable to allow for even shorter time series as neighbors. Naturally, we also allow for neighbors that might not have recent data. A matching with such time series can still be reasonable, due to seasonality effects. Also, not taking such time series into consideration might remove important information from the dataset.

Next, as mentioned in Appendix A.2, we compute the neighborhood's averaging time series on centered sequences $z^c, z \in \overline{\mathcal{N}}_y$, given by $\text{avg}(y) := \text{aDBA}(\{z^c : z \in \overline{\mathcal{N}}_y\})$. As the name already suggests, we now take the averaging time series $\text{avg}(y)$ and map it to $M_{\text{avg}(y)}$.

After estimating model $M_{\text{avg}(y)}$ and its parameters we take this model and use it to forecast on y; i.e. we obtain $\hat{y}_{t+1}(\tilde{M}_{\text{avg}(y)}(y))$. Note that since $M_{\text{avg}(y)}$ is based on centered data, we allow the initial parameters of $M_{\text{avg}(y)}$ to be re-estimated using data of y. This procedure automatically handles the re-centering. This averaging is later referred to as **G-AVG**.

In terms of k-NN, we can write the aggregation function as

$$\begin{aligned}\hat{y}_{t+1} &= f(\bar{\mathcal{N}}(y)) \\ &= \hat{\cdot}(y, \cdot) \circ \tilde{\cdot}(y, \cdot) \circ M \circ \text{aDBA} \circ \text{center}(\bar{\mathcal{N}}(y)),\end{aligned} \tag{3}$$

where, with abuse of notation, we first center the time series of the neighborhood, then average them, model the resulting averaged time series, and finish by refitting and performing the actual forecast with respect to y.

2.3.2. First model, then average approach

As noted above, for a time series $y = (y_s, \dots, y_t)$ we obtain a neighborhood $\mathcal{N}_y$ with size k_y. However, instead of averaging the neighboring time series in terms of DTW, we consider a different approach. As mentioned, each time series $z \in \overline{\mathcal{N}}_y$ has been modeled. That is, there exists M_z for any $z \in \overline{\mathcal{N}}_y$.

In terms of k-NN, the averaging now is of the form

$$\begin{aligned}\hat{y}_{t+1} &= f(\bar{\mathcal{N}}(y)) \\ &= g_w \circ \hat{\cdot}(y, \cdot) \circ \tilde{\cdot}(y, \cdot) \circ M(\bar{\mathcal{N}}(y)),\end{aligned} \tag{4}$$

where g_w is a weighted averaging function, defined as $g_w(x_1, \dots, x_n) := \sum_{i=1}^n w_i x_i$, where n denotes the number of forecasts to be averaged. We have either $n = k_y + 1$ or $n = k_y$. Compared to the G-AVG approach, we clearly see based on Eqs. (3) and (4) that the approaches basically differ in terms of when averaging is performed.

Next, we propose various ways to choose the weights, since it is not clear how to choose them in an optimal way.

2.3.3. Simple average

The easiest and most straightforward way to combine the forecasts is by taking the simple average. We obtain $w_z = w_y = \frac{1}{k_y+1}$. This assumes that all forecasts are equally important. We refer to this type of averaging as **S-AVG**. Alternatively, we can opt not to use M_y but only utilize the models $\tilde{M}_z(y)$ of the neighbors, and set $w_z = 1/k_y$, $w_y = 0$ (**S-AVG-N**).

2.3.4. Distance-weighted average

For the distance-based model average, we consider two ways. First, we can take the DTW distances of the neighborhood and set

$$w_z = \frac{\frac{1}{\delta(y^c, z^c)}}{\sum_{\tilde{z} \in \mathcal{N}_y} \frac{1}{\delta(y^c, \tilde{z}^c)}}, \quad z \ne y, \tag{5}$$

implying that closer neighbors in terms of DTW are assigned higher weights in the model averaging. Since $\delta(y, y) = 0$, we have to set $w_y = 0$ in Eq. (5), implying that this type of averaging only regards neighbors' models. We denote this type of averaging as **D-AVG-N**. This averaging does not take into account the forecasts $\hat{y}_{t+1}(M_y)$ because $w_y = 0$.

In contrast, we can consider the neighborhood's average $\text{avg}(y)$ and corresponding distances $\delta(z^c, \text{avg}(y))$ for $z \in \overline{\mathcal{N}}_y$. Due to the averaging algorithm (see Appendix A.2) and a minimum neighborhood size of 1, we have that $\delta(z^c, \text{avg}(y)) > 0$ for all z. That way, we can set up weights as follows:

$$w_z = \frac{\frac{1}{\delta(z^c, \text{avg}(y))}}{\sum_{z \in \overline{\mathcal{N}}_y} \frac{1}{\delta(z^c, \text{avg}(y))}}, \quad z \in \overline{\mathcal{N}}_y. \tag{6}$$

We denote this type of averaging by **D-AVG**.

2.3.5. Error-weighted average

The error-based or, equivalently, performance-based weights are computed from the models' historical performance. They might be calculated in two ways. First, for each time series $z \in \overline{\mathcal{N}}_y$, we obtain residuals r_z based on

the one-step-ahead forecasts $\hat{z}(M_z)$, i.e. $r_{z,i} := z_i - \hat{z}_i(M_z)$ for $i = s_z + 1, \ldots, t_z$.

Next, we calculate an error measure on the one-step-ahead errors, denoted by $E(z, \hat{z}(M_z))$. Corresponding weights for the model averaging are set to be

$$w_z = \frac{\frac{1}{E(z,\hat{z}(M_z))}}{\sum_{\tilde{z}\in\tilde{\mathcal{N}}_y} \frac{1}{E(\tilde{z},\hat{\tilde{z}}(M_{\tilde{z}}))}}, \quad z \in \overline{\mathcal{N}}_y. \tag{7}$$

This method is denoted by **P-AVG**. In contrast to Eq. (7), we consider the refitted residuals $r_y(\tilde{M}_z(y))$ with $r(\tilde{M}_z(y))_i := y_i - \hat{y}_i(\tilde{M}_z(y))$ for $i = s+1, \ldots, t$ and $z \in \overline{\mathcal{N}}_y$. Consequently, we obtain errors of $E(y, \hat{y}(\tilde{M}_z(y)))$, and the following corresponding weights:

$$w_z = \frac{\frac{1}{E(y,\hat{y}(\tilde{M}_z(y)))}}{\sum_{\tilde{z}\in\tilde{\mathcal{N}}_y} \frac{1}{E(y,\hat{y}(\tilde{M}_{\tilde{z}}(y)))}} \quad z \in \overline{\mathcal{N}}_y. \tag{8}$$

We denote this method by **P-AVG-R**. In practice, we utilize the root mean squared scaled error measure (RMSSE), as introduced in Section 3.

2.3.6. No-model approaches

In respect to the non-parametric neighboring search of k-NN, we also consider non-parametric forecasts as follows. Following Martínez et al. (2017) where forecasts are computed based on the next available observed value, we apply a similar methodology. For the given time series $y = (y_s, \ldots, y_t)$ and neighbors $z = (z_{s_z}, \ldots, z_{t_z}) \in \mathcal{N}_y$, we obtain matchings (on which the neighborhood is based), such that $\phi^{(z)}(k) = (\phi_y(k), \phi_z(k))$ for $k = 1, \ldots, K_z$. Since every index of y must be matched, there exists a value $\tilde{k}$ and $\tilde{t}_z \in \{s_z, \ldots, t_z\}$ such that $\phi^{(z)}(\tilde{k}) = \left(t, \tilde{t}_z\right)$. Next, there are two cases to distinguish.

- Case $\tilde{t}_z < t_z$: Then there exists a successor $u_z = \tilde{t}_z + 1$ with the corresponding value of z_{u_z} which is used to forecast y_{t+1}.
- Case $\tilde{t}_z = t_z$: No successor exists, so the time series z cannot be used to forecast y_{t+1}.

To this end, we obtain the set of possible successors given by

$$\mathcal{S}_y = \left\{ z_{u_z} : u_z = \tilde{t}_z + 1, \tilde{t}_z < t_z, z \in \mathcal{N}_y \right\}. \tag{9}$$

Clearly, Eq. (9) implies that $|\mathcal{S}_y| \leq |\mathcal{N}_y|$. This type of averaging also only considers neighbors' information, because y does not have any successor itself.

As above, we can forecast y_{t+1} by $\hat{y}_{t+1} = \sum_{s\in\mathcal{S}_y} w_s s$ with appropriate weights. We choose simple weights as in $w_s = 1/|\mathcal{S}_y|$ (**S-NM-AVG**), and distance-based weights as in Eq. (5) (**D-NM-AVG**). Note that the matchings are based on centered time series. Hence, the forecasts obtained here are actually forecasts for y^c. Since we do not have a model taking care of re-centering, we have to do it manually by setting $\hat{y}_{t+1} = \hat{y}^c_{t+1} + \bar{y}$.

The methodology's notation is briefly summarized in Table 1.

3. Evaluation methods

For the evaluation of our methods, we use scaled one-step-ahead forecast errors, as proposed by Hyndman and Koehler (2006), adapted to our problem setting of online forecasting. The authors denote the mean absolute scaled error (MASE) by

$$\text{MASE}(y, \hat{y}) = \frac{1}{t-s} \sum_{u=s+1}^{t} |q_u|, \tag{10}$$
$$q_u = \frac{y_u - \hat{y}_u}{\frac{1}{t-s}\sum_{v=s+1}^{t} |y_v - y_{v-1}|},$$

where $y = (y_s, \ldots, y_t)$ is a time series with one-step-ahead forecasts $\hat{y} = (\hat{y}_{s+1|s}, \ldots, \hat{y}_{t|t-1})$. This means that we scale each error by the average in-sample error when using the last available observation as the one-step-ahead forecast, also known as the random-walk forecast.

As mentioned by Hyndman and Koehler (2006), one advantage of this measure is its independence of scale, making it easier to compare results of different time series. Since the measure compares the actual forecast with the mean forecast error based on a random-walk forecast, we can say that if MASE < 1, then the forecast method used to obtain $\hat{y}$ works better on average than the naive approach of using the last available value. Similarly, if MASE > 1, then the method performs worse than the random-walk forecast.

However, in an online forecasting setting, scaling by the whole in-sample error may not be reasonable. Hence, we use a different scaling. We adapt Eq. (10) to a root mean squared scaled error (RMSSE) given by

$$\text{RMSSE}^{t'}_{s'}(y, \hat{y}, \hat{y}^b) := \sqrt{\frac{1}{t'-s'+1}\sum_{u=s'}^{t'} q_u^2}, \tag{11}$$
$$q_u = \frac{y_u - \hat{y}_u}{\sqrt{\frac{1}{u-s}\sum_{v=s+1}^{u}(y_v - \hat{y}^b_v)^2}},$$

where $s \leq s' \leq t' \leq t$. This means we obtain scaled errors q_u by scaling the error $y_u - \hat{y}_u$ by the averaged benchmark forecast error until time u. Altogether, $\text{RMSSE}^{t'}_{s'}$ in Eq. (11) gives the average error of the window $\{s', \ldots, t'\}$ scaled by the average historical benchmark forecast until time t'. A typical benchmark method is the random-walk forecast given by $\hat{y}^b_v = y_{v-1}$.

As the data are usually split into training and test sets, we also need to differentiate the corresponding evaluation techniques. In the training set, we compute the RMSSE as given above, yielding a final RMSSE value for the entire training set.

For an evaluation on the test set, we follow the notation of Hyndman and Koehler (2006) and its use in the R package **forecast** (Hyndman & Khandakar, 2008) where the test errors are scaled by the training error of the random walk. Consider a time series $y = (y_s, \ldots, y_t, y_{t+1}, \ldots, y_T)$ split in training and test sets at time step t. Then, each test set residual is scaled by the root mean squared

Table 1
Averaging notation.

Avg. Notation	Description
G-AVG	Forecast the averaged time series
S-AVG	Model Avg. using simple weights
S-AVG-N	Model Avg. using simple weights and only neighbors
D-AVG	Model Avg. using distance-based weights
D-AVG-N	Model Avg. using distance-based weights and only neighbors
P-AVG	Model Avg. using performance-based weights
P-AVG-R	Model Avg. using refitted performance-based weights
S-NM-AVG	No Model Avg. using simple weights
D-NM-AVG	No Model Avg. using distance-based weights

Table 2
Summary of datasets.

Dataset name	No. of time series	Forecast horizon	Mean length	Length range
Food Demand	43	1–4	33	2 – 78
M3	1,376	18	119	66 – 144
Tourism	366	24	299	91 – 333
CIF 2016	72	6, 12	99	28 – 120
Hospital	767	12	84	84 – 84

error of the random-walk forecast yielding scaled test set residuals given by Eq. (12).

$$q_u = \frac{r_u}{\sqrt{\frac{1}{t-s}\sum_{v=s+1}^{t}(y_v - \hat{y}_v^b)^2}}, \quad T \geq u > t. \tag{12}$$

In addition to the RMSSE, we consider more common forecasting measures, such as the mean absolute error (MAE) (13), root mean squared error (RMSE) (14), and symmetric mean absolute percentage error (sMAPE) (15); see Hyndman and Koehler (2006) for a recent review.

$$\text{MAE} = \frac{\sum_{i=t+1}^{T}|y_i - \hat{y}_i|}{T-t} \tag{13}$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=t+1}^{T}(y_i - \hat{y}_i)^2}{T-t}} \tag{14}$$

$$\text{sMAPE} = \frac{2}{T-t}\sum_{i=t+1}^{T}\frac{|y_i - \hat{y}_i|}{|y_i| + |\hat{y}_i|} \tag{15}$$

For each dataset the mean and median error for each measure is computed.

4. Experiments

To demonstrate the methodology, we use five publicly available datasets. Table 2 provides summary information of these datasets.

All datasets are on a monthly basis, except for the Food Demand data, which is a weekly dataset. A short summary is as follows.

- Food Demand. Weekly data from smart fridges with the goal to forecast the demand of each fridge for the upcoming week in a one-step-ahead fashion. The dataset is available in this paper's R package.
- M3 Dataset (Makridakis & Hibon, 2000). Monthly data of the M3 competition, split in five subcategories: micro, macro, industry, demographic, and finance. The sixth subcategory (called 'other') was removed from the analysis. We use this prior domain knowledge and model each subcategory individually. The data were obtained from the R package **Mcomp** (Hyndman, 2018).
- Tourism (Athanasopoulos et al., 2011). Monthly dataset from the tourism forecast competition. The data are available from the R package **Tcomp** (Ellis, 2018).
- CIF 2016 (Stepnicka & Burda, 2017). Monthly data from the CIF 2016 forecasting competition. The data contain 24 real-time series from the banking domain and 48 artificially created time series. The dataset is available from Zenodo (Godahewa et al., 2020) and is put in this paper's corresponding R package for convenience.
- Hospital. Monthly time series counting patients for different medical products and related medical problems. The dataset is available in the R package **expsmooth** (Hyndman, 2015).

All datasets in Table 2 have a forecast horizon greater than one. The Food Demand dataset is a special case, since we are only interested in one-step-ahead forecasts for that dataset. Hence, we run two experiments on these datasets: multi-step-ahead forecasts as required for the datasets, with cumulative one-step-ahead forecasts for a comparison to the Food Demand data results. We do not compute multi-step forecasts for the Food Demand data.

We denote by TSAVG the trained averaging method. For all datasets, the same training procedure was performed.

4.1. Baseline model

For the base family of models, any common time series model family can be used. We here use the ETS family of models (Hyndman et al., 2002; Taylor, 2003). These models consist of three components: **E**rror, **T**rend, and **S**easonality.

The most simple ETS model is of the form *ANN* and is also known as exponential smoothing. It does not have

any trend or seasonality component, and is given by the recursion

$$\begin{aligned} l_t &= \alpha y_t + (1-\alpha) l_{t-1}, \\ \hat{y}_{t+h|t} &= l_t, \quad h > 0, \end{aligned} \tag{16}$$

where l denotes the level component of the model, which is also equal to the flat forecast $\hat{y}_{t+1|t}$. The model parameter α is usually found by minimizing the sum of squared one-step-ahead forecast errors.

The next, more complex model is of the form *AAN* and is called the Holt–Winters model. It is given by

$$\begin{aligned} l_t &= \alpha y_t + (1-\alpha)(l_{t-1} + b_{t-1}), \\ b_t &= \beta(l_t - l_{t-1}) + (1-\beta) b_{t-1}, \\ \hat{y}_{t+h|t} &= l_t + h b_t, \quad h > 0, \end{aligned} \tag{17}$$

where l again denotes the level component. The newly introduced trend component is denoted by b. In this model, the forecasts are not flat anymore, but linear in the forecast horizon h. All smoothing parameters are usually constrained between 0 and 1. For a more detailed review of those models and many more equations like Eqs. (16) and (17), see Gardner (1985).

With that, this family provides a very flexible way of modeling and forecasting, and is therefore often used in business forecasting applications. The **forecast** package (Hyndman & Khandakar, 2008) in R also provides an automatic forecasting framework for ETS and other models (such as ARIMA), where the most appropriate model of all possible models in that family is chosen automatically based on some optimization criteria. We use the corrected Akaike information criterion, given by

$$\mathrm{AIC}_c = \mathrm{AIC} + \frac{2k(k+1)}{T-k-1} = -2\log(L) + 2k + \frac{2k(k+1)}{T-k-1}, \tag{18}$$

where L is the model's likelihood, k denotes the total number of parameters in the model, and T is the sample size. The correction is needed because in small samples the regular Akaike information criterion (AIC) tends to overfit and select models with too many parameters (Hurvich & Tsai, 1989).

Each time series is modeled by ETS and then forecasted in a one-step-ahead fashion, regardless of its length.

Another common model family of time series models in ARIMA (Shumway & Stoffer, 2000) is used to model the dataset, again utilizing the **forecast** package. As for the ETS family, the best model is chosen by the corrected AIC as defined in Eq. (18).

4.2. Global benchmark model

As a second benchmark model we use a pooled autoregressive (AR) model (Baltagi, 2021), which can be seen as a global model, since it uses all individuals' information. This linear $AR(p)$ model is given by

$$y_{i,t} = \alpha + \sum_{k=1}^{p} \beta_k y_{i,t-k} + \epsilon_{i,t}, \tag{19}$$

where $i = 1, \ldots, N$ denotes the ith individual, and $t = s_i, \ldots, t_i$ denotes the time component of the panel. The order of the model is given by $p \geq 1$. This means that for the entire panel data we only need to estimate $p+1$ parameters in Eq. (19) – the global intercept α as well as the global slope parameters $\beta_1, \ldots, \beta_p$. In practice, the observations of all individuals are stacked on top of each other to obtain a simple linear model which can be estimated by ordinary least squares.

Such models, also called panel models, can be of different forms as well. We also modeled the data using variable intercepts; i.e. each individual has an individual intercept. However, this model turned out to be worse than the pooled model for the datasets.

Another variant is the variable-coefficient model with individual intercept and slope. However, such a model is usually estimated by estimating each individual's model by its own. Thus, such an approach cannot really be seen as a global model, so we opted not to use this model as well.

For more details about the models and their statistical properties, see the book by Baltagi (2021). All global panel models were fitted using the **plm** package (Croissant & Millo, 2008) in R. Lag selection for $p = 1, \ldots, L_{\max}$ was performed using time series cross-validation, as in Section 4.3. For each dataset an individual maximum lag was selected based on the rule of thumb of Bandara et al. (2020), namely input size $= 1.25 \max(\text{output size}, \text{seasonal period})$. The specific values of $L_{\max}$ are given in Table 3, and five equidistant values for p are chosen for each dataset.

4.3. Choice of optimal parameter

In every experiment, we perform time series cross-validation (TSCV) based on a rolling window approach to choose the optimal hyperparameter k in k-NN. We cannot apply regular cross-validation, since the assumption of independent data is not valid in a time series context.

The initial window contains observations from time step 1 until time step T_0. In each iteration, the window is expanded by one time step, yielding a new fold. In each fold, we obtain an RMSSE value for each individual and hyperparameter. In detail, we assume that there exists a hyperparameter $k \in \Theta$, where Θ is the search grid. Let a fold be $f = \{1, \ldots, T_0, \ldots, t_f\}$, $T_0 \leq t_f \leq T_{\text{train}}$ and consider an individual time series $y^{(i)} = (y_{s_i}, \ldots, y_{t_i})$. Then, we compute $\mathrm{RMSSE}^{t_f}_{\max(T_0,s_i)}(y^{(i)}, \hat{y}^{(i)}(m); k)$ with random-walk benchmark forecasts for each parameter k, time step t_f, individual i, and method m. Furthermore, the cross-validation yields standard errors as well, by aggregating over the folds:

$$\mathrm{SE}(y^{(i)}, \hat{y}^{(i)}; k) = \sqrt{\frac{1}{T_{\text{train}} - \max(T_0, s_i) + 1}} \tag{20}$$
$$\sqrt{\frac{\sum_{t_f=\max(T_0,s_i)}^{T_{\text{train}}} \left(\mathrm{RMSSE}^{t_f}_{\max(T_0,s_i)}(y^{(i)}, \hat{y}^{(i)}(m); k) - \mu(y^{(i)}, \hat{y}^{(i)}(m); k)\right)^2}{T_{\text{train}} - \max(T_0, s_i)}},$$

with a mean cross-validation error, also known as the CV score, of

$$\mu(y^{(i)}, \hat{y}^{(i)}(m); k) = \frac{\sum_{t_f=\max(T_0,s_i)}^{T_{\text{train}}} \mathrm{RMSSE}^{t_f}_{\max(T_0,s_i)}(y^{(i)}, \hat{y}^{(i)}(m); k)}{T_{\text{train}} - \max(T_0, s_i)}. \tag{21}$$

Table 3
Summary of training parameters. T_0 denotes the initial training timestep, T_{train} the final training timestep and $L_{\max}$ the maximum lag in the pooled model.

Dataset Name	T_0	mean T_{train}	$L_{\max}$
Food Demand	21	80	5
M3 - Demographic	92	308	22
M3 - Finance	94	411	22
M3 - Industry	63	172	22
M3 - Macro	73	378	22
M3 - Micro	87	111	22
Tourism	131	316	30
CIF 2016	96	109	15
Hospital	58	72	15

Note here that each individual i might be available in a different amount of folds, since we perform cross-validation based on the time aspect of the panel data. The optimal hyperparameter is now chosen using the one-standard-error rule. That is, we choose the most parsimonious model lying in the one-standard-error band of the global minimum of cross-validation errors values. This means that based on Eqs. (20) and (21), we obtain the optimal parameter k^* for $y^{(i)}$, as in the minimization problem (22), where $k_i^{(m)}$ is the parameter realizing the minimal mean cross-validation error, i.e. $k_i^{(m)} = \text{argmin}_k \mu(y^{(i)}, \hat{y}^{(i)}(m); k)$.

$$k_i^* = \min\left\{\theta : |k - k_i^{(m)}| \leq \text{SE}(y^{(i)}, \hat{y}^{(i)}; k_i^{(m)})\right\}. \tag{22}$$

Training parameters such as T_0 are all listed in Table 3 for each dataset. The reason why a mean value is given for T_{train} is because of the possibly different lengths of the time series.

4.4. Results

Tables 4, 5, 6, and 7 show the corresponding mean and median errors for the multi-step-horizon case for each dataset.

First, we note that the pooled AR model seems to perform well for the M3 and Tourism datasets. Those datasets are characterized by long forecast horizons, as well as long time series, which could be a reason for the good performance. When comparing the averaging methodology to the local benchmarks ETS and ARIMA, there is no clear picture present. Nevertheless, TSAVG does seem to keep up with the benchmark methods and is able to improve the performance of the ETS model, which it is supposed to do. For testing statistical significance, non-parametric Friedman tests are performed on the mean values, and the corresponding p-values are given in the table captions. No statistical significant difference for an α-level of 5% is found for any error measure.

The corresponding one-step-ahead errors are as seen in Tables 8, 9, 10, and 11. We observe a similar picture: the pooled model performs well for the M3 and Tourism datasets. When comparing TSAVG to ETS and ARIMA, the numbers are very close and no clear best method is present. In the case of one-step-ahead errors, TSAVG is only able to improve the performance of the ETS model in a few cases. In the case of the Food Demand dataset (containing just a small number of rather short time series) we observe improvements over the local base models as well as all other competing models. As before, Friedman tests are performed on the mean values. The null hypothesis of no differences cannot be rejected for any error measure again.

To summarize the results, the proposed method is competitive on all datasets. On the Food Demand dataset, we observe especially good performance due to its characteristics. However, TSAVG also performs comparably on datasets which do not share the same properties as the Food Demand dataset. For example, the Hospital dataset only contains equally sized time series, whereas the Tourism dataset contains very long time series. This makes TSAVG a very flexible method to improve simple local models for a wide range of datasets.

When looking at all results with respect to the size of the datasets, there does not seem to be a huge impact. We do, however, observe that the bigger the dataset, the better TSAVG seems to perform compared to the baseline ETS model. This is reasonable, since the number of possible neighbors increases with increasing dataset size.

4.5. Runtime

Fig. 2 shows training runtimes for a selection of averaging methods. We clearly see the effect of increasing the neighborhood size across all datasets and averaging methods. The runtime also increases significantly with the dataset size, due to the pairwise search for DTW neighbors. While the calculation times of P-AVG and S-AVG are similar because the corresponding weights are both quickly computed, the distance-based averaging method has increasing computation time because it needs to compute the neighborhood's averaged time series.

4.6. Diagnostics for the food demand data

The transparency of TSAVG allows for advanced diagnostics. These are demonstrated on the Food Demand dataset, which is about forecasting weekly demand of smart fridges. Its rather small size and short time series lead to understandable diagnostics on an individual time series level. The practical problem is also what initially motivated the idea of using pattern matching to improve forecasts. A selection of example time series of the dataset are available in Fig. 3 and show its heterogeneity. While homogeneity is still present, individual 29, for example, shows a very similar pattern to the first part of individual 3. This is the similarity we ought to extract using DTW.

4.6.1. TSCV diagnostics

In each fold, we forecast the time series to obtain one-step-ahead forecasts, as explained in Section 4.3. The ETS models themselves are not being tuned using the folds. The best ETS model is solely chosen in each fold using the AIC. That way, the only tuning parameter is the size of each individual's neighborhood.

We take a closer look at the TSCV procedure and the choice of the optimal tuning parameters. Fig. 4 shows

Table 4
Multi-step MAE (p-value 0.127).

Model	M3		CIF2016		HOSPITAL		TOURISM	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
TSAVG	713.580	510.021	**545 365.6**	98.162	21.440	**6.763**	7109.038	985.100
ETS	736.036	507.657	642 448.9	**90.546**	22.553	6.782	7391.902	959.128
ARIMA	704.468	487.471	568 230.7	105.003	20.665	6.869	4184.401	811.594
POOL	**678.815**	**485.306**	1300457.8	219 661.126	**20.196**	7.248	**2258.485**	**501.032**

The best results are in bold.

Table 5
Multi-step RMSE (p-value 0.127).

Model	M3		CIF2016		HOSPITAL		TOURISM	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
TSAVG	860.793	604.199	**632 468.5**	117.239	26.225	8.347	8632.070	1241.053
ETS	886.086	626.804	722 426.3	**107.687**	27.336	8.324	8974.416	1222.899
ARIMA	845.835	**588.559**	641 597.5	126.505	25.021	**8.241**	5377.097	1090.481
POOL	**815.725**	592.032	1456160.9	249 604.091	**24.309**	8.709	**2743.846**	**601.960**

The best results are in bold.

Table 6
Multi-step RMSSE (p-value 0.44).

Model	M3		CIF2016		HOSPITAL		TOURISM	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
TSAVG	**1.915**	1.178	1.121	0.864	0.878	**0.822**	1.761	1.468
ETS	1.967	1.220	**1.105**	**0.809**	0.900	0.828	1.853	1.497
ARIMA	1.927	**1.108**	1.191	0.896	**0.871**	0.822	1.597	1.414
POOL	2.293	1.109	105 106.814	2050.317	0.907	0.856	**1.327**	**0.934**

The best results are in bold.

Table 7
Multi-step sMAPE (p-value 0.825).

Model	M3		CIF2016		HOSPITAL		TOURISM	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
TSAVG	0.159	0.104	0.158	**0.079**	0.181	0.168	0.352	0.295
ETS	0.164	0.106	**0.144**	0.082	0.185	0.166	0.369	0.306
ARIMA	0.161	**0.101**	0.169	0.091	**0.179**	**0.166**	0.325	0.283
POOL	**0.153**	0.104	1.589	1.883	0.190	0.173	**0.311**	**0.187**

The best results are in bold.

Table 8
Cumulative one-step ahead MAE (p-value 0.564).

Model	M3		CIF2016		HOSPITAL		TOURISM		FOOD	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median
TSAVG	484.660	322.573	301 584.2	82.064	19.610	**6.437**	3647.229	770.753	**8.523**	**6.917**
ETS	478.813	318.437	247 264.0	**76.378**	19.635	6.491	3744.562	767.282	8.662	7.824
ARIMA	**470.654**	**311.126**	**233 306.1**	79.007	19.040	6.495	3190.488	700.724	9.882	9.720
POOL	475.890	313.683	411 428.1	32 944.310	**17.643**	6.854	**1543.199**	**461.658**	9.438	7.387

The best results are in bold.

Table 9
Cumulative one-step ahead RMSE (p-value 0.564).

Model	M3		CIF2016		HOSPITAL		TOURISM		FOOD	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median
TSAVG	615.987	406.703	360 462.5	**93.481**	23.985	8.080	4759.500	1005.795	**10.112**	**8.417**
ETS	611.354	410.237	305 246.7	93.493	23.991	**7.992**	4870.787	1008.100	10.193	8.720
ARIMA	598.826	395.709	**301 763.1**	96.834	23.320	8.022	4119.036	929.144	11.552	11.365
POOL	**597.438**	**389.837**	451 571.8	32 944.598	**21.610**	8.357	**2047.617**	**597.842**	11.526	9.685

The best results are in bold.

the corresponding TSCV scores and standard errors, as defined above for the distance-based averaging methods. Each plot panel shows the results for a selected smart fridge. For demonstration purposes, we choose a grid of

Table 10
Cumulative one-step ahead RMSSE (p-value 0.241).

Model	M3		CIF2016		HOSPITAL		TOURISM		FOOD	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median
TSAVG	0.904	0.805	0.885	0.802	0.824	0.800	1.418	1.320	**0.875**	**0.748**
ETS	0.892	0.796	**0.759**	0.751	0.823	0.802	1.423	1.348	0.887	0.762
ARIMA	**0.882**	**0.780**	0.770	**0.738**	**0.819**	**0.796**	1.282	1.205	0.973	0.804
POOL	1.110	0.839	25 562.026	280.377	0.842	0.824	**0.929**	**0.816**	1.354	0.939

The best results are in bold.

Table 11
Cumulative one-step ahead SMAPE (p-value 0.948).

Model	M3		CIF2016		HOSPITAL		TOURISM		FOOD	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median
TSAVG	0.115	0.069	0.107	0.084	**0.168**	**0.158**	0.291	0.262	**0.244**	**0.211**
ETS	0.114	0.068	**0.097**	0.082	0.169	0.160	0.288	0.265	0.251	0.214
ARIMA	0.114	**0.065**	0.101	**0.080**	0.169	0.161	0.267	0.241	0.301	0.242
POOL	**0.114**	0.066	1.338	1.592	0.176	0.169	**0.206**	**0.158**	0.274	0.230

The best results are in bold.

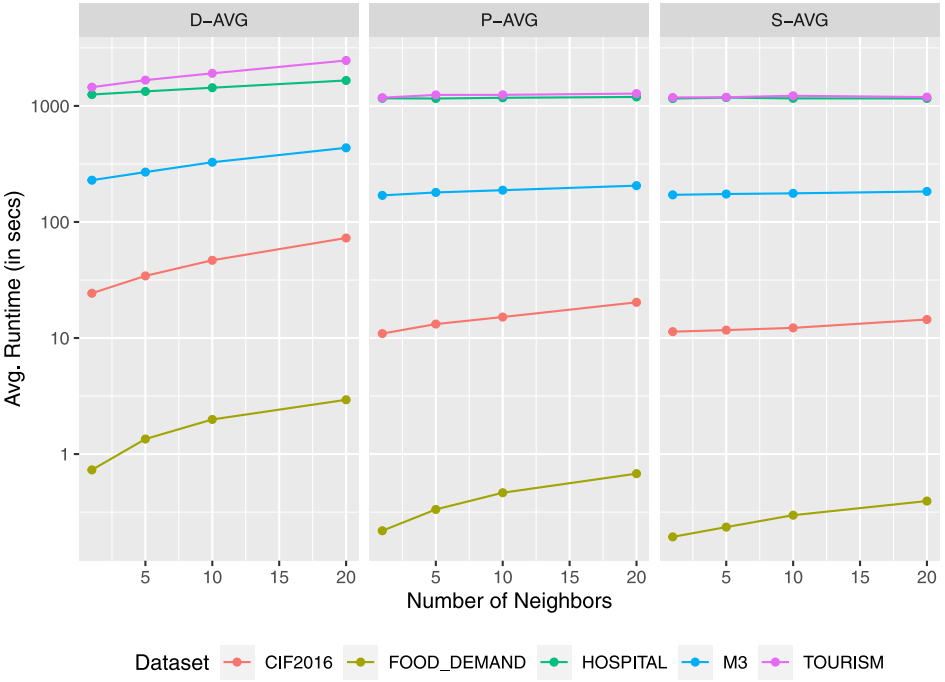


Fig. 2. Runtime analysis for each dataset and selected averaging methods. The numbers shown are averaged runtimes of the TSCV procedure.

$\Theta = \{1, 3, 5, 10, 20\}$ number of neighbors and select the optimal number by applying the one-standard-error rule. We do this for each forecast averaging methodology. The vertical lines indicate the choice of the tuning parameters. We observe different behavior between the global averaging approach and the others (D-AVG and D-AVG-N). This can be explained by the similarity of the two methods. We also see that the scores for individual 3 obtain a constant level because this individual is one of the longer time series, and hence it is not even possible to have more than three neighbors.

The actual selected number of neighbors is given in Table 12. See Table 1 for a description of all averaging approaches.

4.6.2. Model evaluation

First, we evaluate the experiment on a global level. This is where the scaled errors come in handy, since these are also comparable on an individual level. Fig. 5 shows boxplots of the RMSSE values, as in Eq. (11), split in both training and test errors. Given that the errors are skewed, we display them on a log scale. We clearly

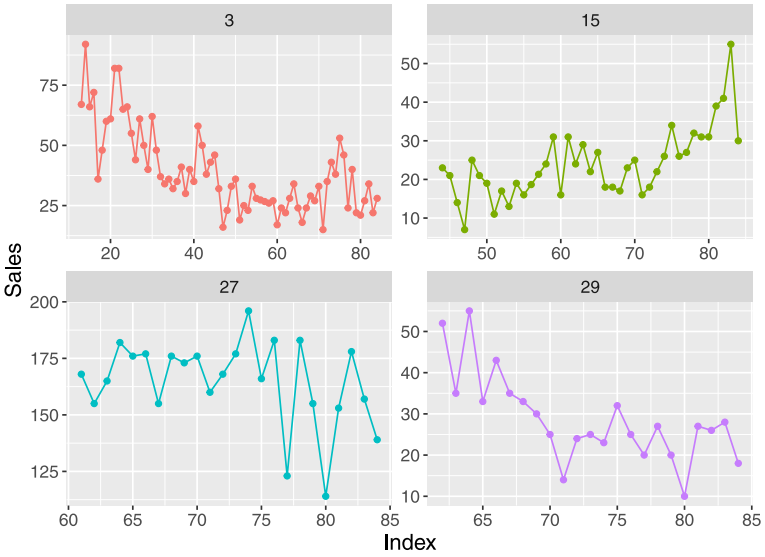


Fig. 3. Example time series of the Food Demand dataset.

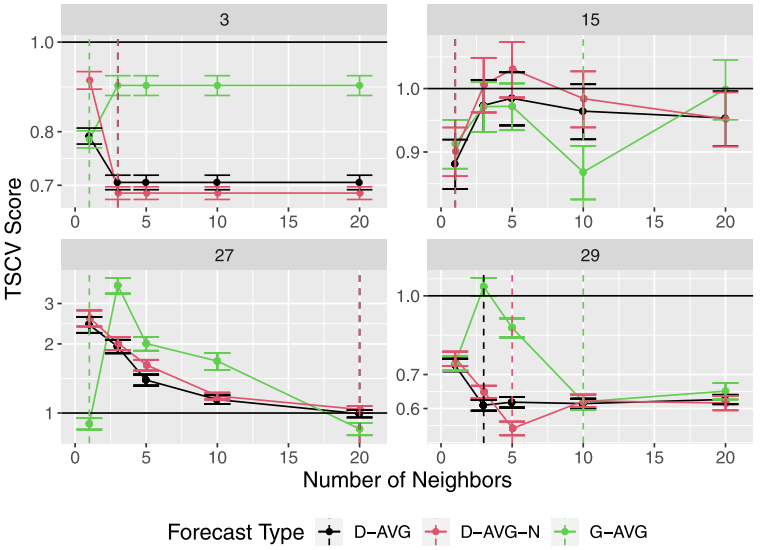


Fig. 4. TSCV scores for distance-based averaging. The dashed vertical lines indicate the optimal number of neighbors.

observe that the simple, non-models-based averaging approaches yield worse results than the ETS benchmark model. However, we do see minor improvements in the ETS forecasts from using distance- and performance-based averaging methods. Another interest fact is that approaches that do not consider an individual's information (S-AVG-N and D-AVG-N) also tend to yield worse results, indicating that this information should also be included in the forecast procedure, even though these time series might be very short and difficult to forecast. For completeness, Fig. C.15 in the appendix shows the original RMSSE values, but as mentioned above, this makes comparisons to the ETS benchmark model quite difficult.

Next, we dig deeper into the performance of selected individuals. For this, we selected four of the 43 individuals to analyze in more detail. Fig. 6 shows the RMSSE with respect to ETS forecasts for these selected fridges. We also show standard errors of the RMSSE. Due to the different behavior of the individuals, we also observe the different performance of the averaging approaches. While for individual 3, we can improve forecasts using model-based averaging methods, this is not the case for individual 27. We also observe that the panel benchmark model PLM-POOL does not always yield better forecasts and is often even surpassed by any of the averaging approaches.

We can dig even deeper into evaluating the errors when looking at the evolution of the RMSSE. For reasons

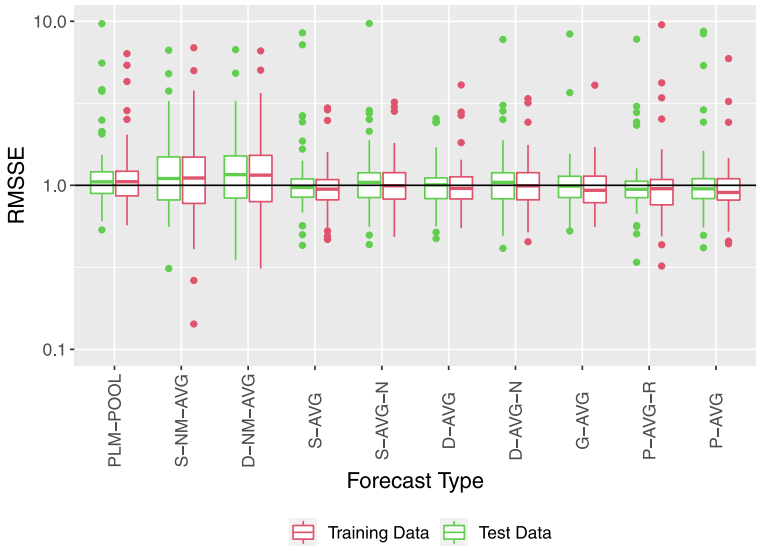


Fig. 5. Global RMSSE with respect to ETS forecasts.

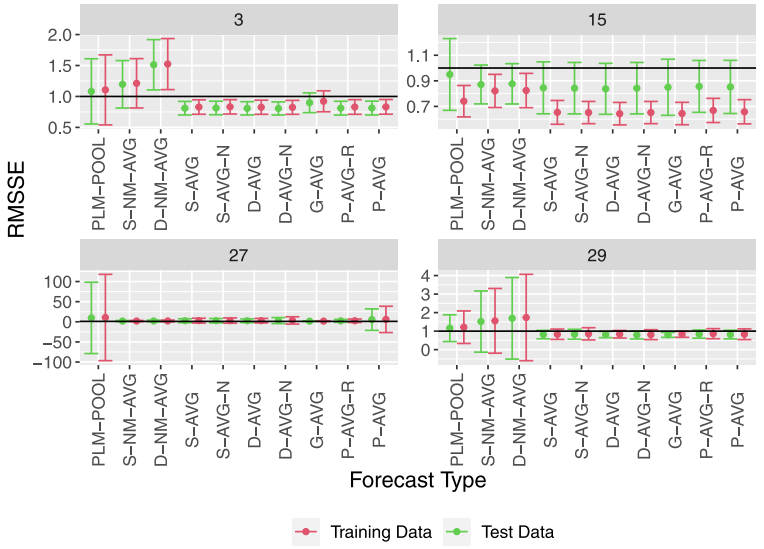


Fig. 6. Individual RMSSE with respect to ETS forecasts.

of clarity, Fig. 7 shows the running RMSSE with respect to the random-walk forecasts for the distance-based averaging methods and the two benchmark methods of ETS and PLM-POOL. The dashed vertical line indicates the split between training and test periods. These plots really show the possibility of improving the benchmark forecasts by smartly taking averages of neighbors' forecasts. Similar plots for the remaining averaging methods can be found in the appendix.

4.6.3. Further diagnostics

To better understand the differences in performance on an individual level, we perform further diagnostics. For a fixed neighborhood size of at most five neighbors, we first consider the median, minimum, and maximum normalized DTW distances and their evolution over time. This analysis might already give hints on the homogeneity of the neighborhoods (Fig. 8). We observe that for individuals 3, 15, and 29, the distances and hence the neighborhoods seem to stabilize after the first few steps, while for individual 27, the normalized distance increases until the end. This means that this individual becomes harder and harder to match. Overall, individual 27 is too heterogeneous and lacks homogeneity to improve forecasts as also indicated by the high mean 2-Wasserstein distance in Fig. 10.

Another way to evaluate the neighborhoods is as follows. We consider the final training neighborhoods as

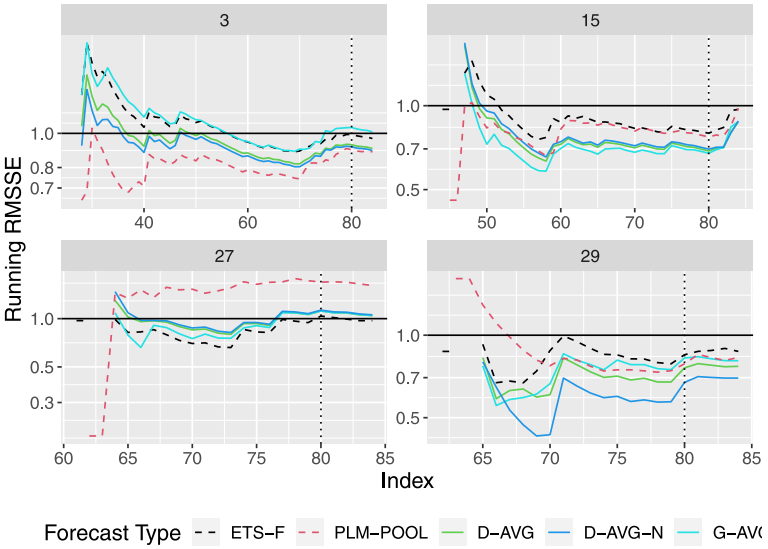


Fig. 7. Running RMSSE with respect to random-walk forecasts for distance-based averaging. The dotted lines indicate the split between training and test set.

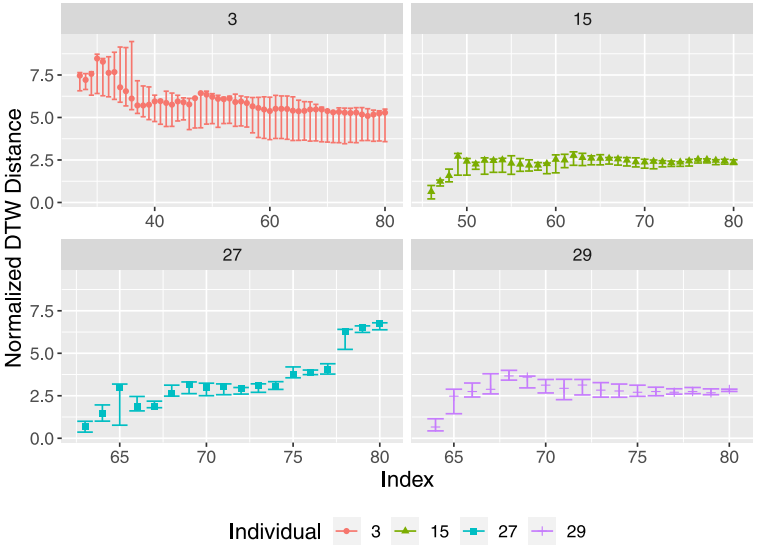


Fig. 8. Normalized DTW distance evolution over time.

the ground truth and compare each neighborhood yielded in the training process to this ground truth using the adjusted Rand index (Rand (1971), Hubert and Arabie (1985)). Fig. 9 shows the adjusted Rand index for each time step. We see that individual 3 is perfectly matched from the beginning. This is unsurprising because this individual has few possible neighbors. However, there is much more variability present for the remaining smart fridges. In particular, for individual 27, the adjusted Rand index hardly increases, indicating difficulties in finding the proper neighbors for this individual.

Finally, we look at the 2-Wasserstein distances, as discussed in Appendix A.3. Since we only introduced the Wasserstein for a basic *ANN* model, we go into more detail here. The one-step-ahead point forecasts which we obtain and use for averaging are the means of the forecast distribution. Hence, we can compute the Wasserstein distances in a straightforward way. Specifically, we have that

$$W_2^2(\hat{X}_{n+1|n}, \hat{Y}_{n+1|n}) = (\hat{x}_{n+1|n} - \hat{y}_{n+1|n})^2 + (\hat{\sigma}_X - \hat{\sigma}_Y)^2, \quad (23)$$

where X, Y are two arbitrary ETS models with realized point forecasts $\hat{x}, \hat{y}$ as well as estimated standard deviations $\hat{\sigma}_X, \hat{\sigma}_Y$, respectively.

Fig. 10 shows the mean 2-Wasserstein distances, as in Eq. (23), per selected individual. The neighborhoods considered here are the same as in Fig. 8. The dashed lines

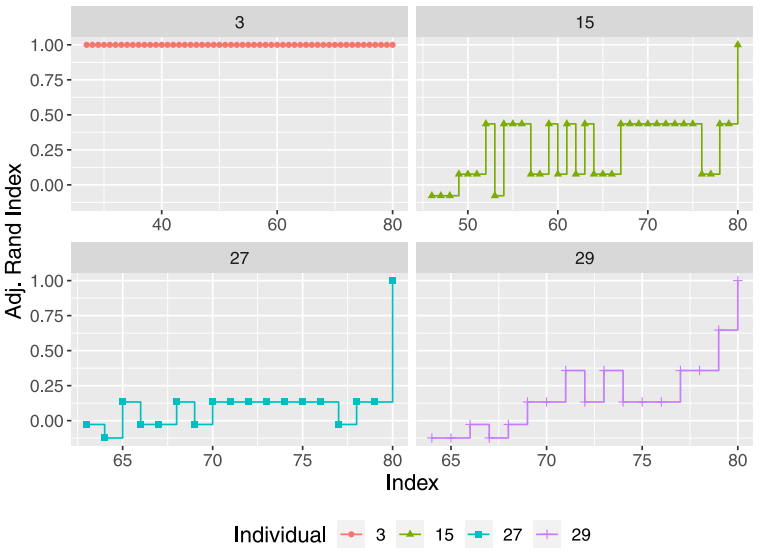


Fig. 9. Adjusted Rand index over time.

Table 12
Optimal neighborhood sizes based on TSCV.

Forecast type	Individuals			
	3	15	27	29
S-NM-AVG	1	5	20	20
D-NM-AVG	3	5	20	10
S-AVG	3	1	20	5
S-AVG-N	3	1	20	5
D-AVG	3	1	20	3
D-AVG-N	3	1	20	5
G-AVG	1	10	1	10
P-AVG-R	3	1	10	5
P-AVG	3	1	20	5

indicate the overall mean distance. We clearly observe large distances for individual 27, which is consistent with the rather bad results. As in Fig. 8, the DTW distances are increasing for this individual. As such, we cannot expect the Wasserstein distances to be small and thus cannot guarantee a reduction in forecast error. Indeed, the overall mean distance is 12.5, more than twice as large as the mean values for the remaining individuals (3.2 for individuals 3 and 15, and around 5.2 for individual 29.) selected in the analysis. This result empirically verifies the extension of our motivating theoretical results.

4.6.4. The best model

By minimizing the mean RMSSE values, we can choose the optimal averaging technique on the dataset. With a mean RMSSE value with respect to the ETS base forecast of 1.08, the optimal averaging method is P-AVG-R. This rather high value is due to the bad performance of a small amount of individuals, skewing the mean value. The median values account to 0.85. Table 13 shows the results in terms of a number of error measures. While for individual 3, the averaging yields uniformly better results, this is not the case for individual 15, where the base ETS model and the pooled global model outperform the averaging methodology.

To test statistical significance, we follow the suggestions by Demšar (2006). A non-parametric Friedman test is performed at an α-level of 5% for each error measure using the results for all individuals. This yields significant differences for all error measures ($p < 0.01$). Post hoc pairwise Wilcoxon tests with Holm's α-level correction are then performed. However, these tests do not yield any significant pairwise differences in the methods for any error measure.

Additionally, we take a look at the distributional properties of the RMSSE values obtained on the test set and compare the methods based on how many individuals could be improved by averaging. First, Fig. 11 shows the ordered RMSSE test set values with respect to the ETS forecast for each averaging method. The vertical lines indicate the percentile of individuals with a smaller error than the ETS benchmark. For an overall useful methodology we want the vertical lines to be greater than 0.5, meaning that we can improve the forecasting for more than 50% of the individuals. This is the case for the global averaging method G-AVG, the performance-based P-AVG and P-AVG-R, and the simple S-AVG, whereby the best improvement is obtained by the weighted average of using errors of the refitted models (P-AVG-R). Furthermore, we see that all methods not including the individual, namely D-AVG-N and S-AVG-N, yield worse results. Also, the non-model-based methods which only regard the DTW matchings do not seem to be very useful at all, since they only decrease the forecast error compared to the ETS benchmark for around 36% and 38% of the individuals.

5. Conclusions

We presented a general framework for the improvement of individual forecasts in a set of possibly

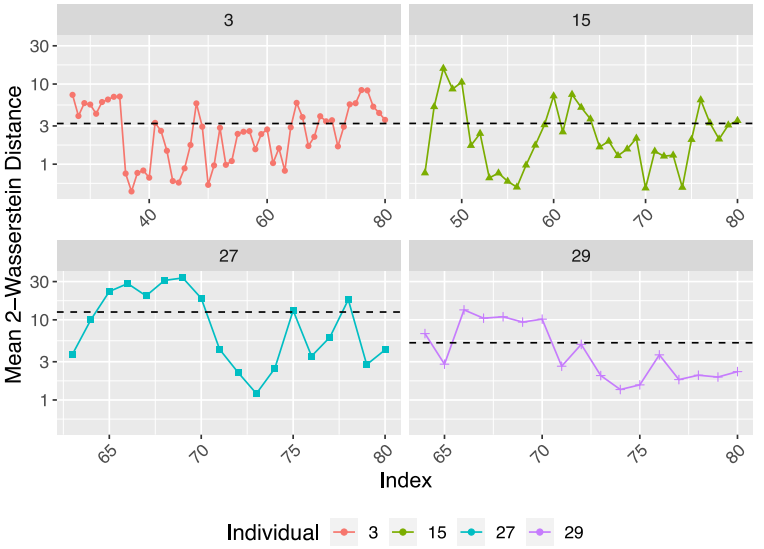


Fig. 10. Wasserstein distances of the forecast distributions.

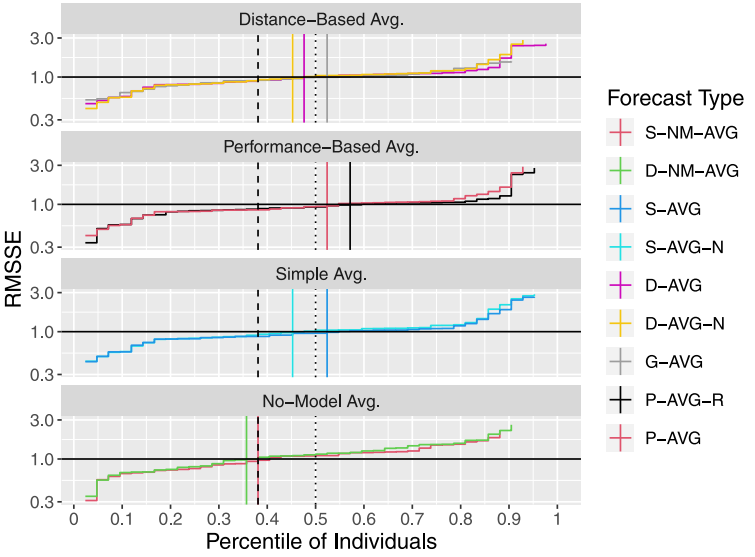


Fig. 11. Percentile plots for each averaging method with respect to ETS forecasts. The dashed lines show the performance of the global panel benchmark model. The dotted lines indicate the 50% percentile.

Table 13
Test errors for benchmark models and best averaging method (P-AVG-R).

Model	sMAPE				MAE				RMSE				RMSSE			
	3	15	27	29	3	15	27	29	3	15	27	29	3	15	27	29
TSAVG	**0.147**	0.312	**0.083**	0.296	**4.094**	12.184	**13.177**	6.699	**5.353**	12.946	**16.033**	**6.962**	**0.439**	1.901	**0.609**	**0.668**
ETS	0.163	0.298	0.097	**0.294**	4.531	**11.656**	15.466	**6.566**	5.559	**12.321**	16.883	7.103	0.456	**1.809**	0.642	0.682
ARIMA	0.222	0.332	0.097	0.367	6.127	13.299	15.467	7.918	6.765	14.247	16.885	8.832	0.554	2.092	0.642	0.848
POOL	0.225	**0.297**	0.169	0.310	5.073	12.097	22.788	7.160	6.873	12.890	30.948	7.941	0.595	1.870	1.124	0.893

The best models are in bold.

heterogeneous time series. It is based on a theoretical motivation regarding simple state-space models in terms of exponential smoothing. However, this motivation may be extended to more general models. In the theoretical part of the work, new approaches were proposed by computing the DTW distance explicitly on random processes,

as well as using the Wasserstein distance to compare the forecast distribution of state-space models. Additionally, dynamic time warping was introduced and described in a far more rigorous way.

We applied a sequence matching procedure in dynamic time warping, which allows us to find similar time series in terms of their shape and which works for any two time series. To this end, we emphasized the use of asymmetric matching and the corresponding extension of the global averaging methodology to this type of matching. The transparency of the algorithm also enables us to perform diagnostics and to understand when this procedure is appropriate and yields reasonable results. This point also differentiates our work from machine learning approaches, which tend to be opaque and require a large total number of observations to work efficiently.

The models we used for our analysis were ETS models, yet any family of models can be used in our procedure, making it very flexible in practice. It also extends naturally to automatic forecasting frameworks such as ARIMA or ETS. While the methodology may not be able to improve the performance of the base model all the time, it still yields competitive results. Overall, this framework fits many real-world applications where one encounters a set of heterogeneous and possibly short time series.

Some aspects are still to be considered. One extension of the work could be using an adaptive data-driven number of neighbors for each time series, instead of a fixed one. Further, a robustification of the procedure could be beneficial, since dynamic time warping is quite sensitive to outliers in the time series.

Computational details

The results in this paper were obtained using R 4.1.3. Figures were produced by **ggplot2** (Wickham, 2016), and tables were created using **knitr** (Xie, 2022) and **kableExtra** (Zhu, 2021). R itself and all packages used are available from the Comprehensive R Archive Network (CRAN) at https://CRAN.R-project.org/.

The source code of this paper in form of the R package **TSAvg** is available from GitHub at https://github.com/neubluk/TSAvg.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lukas Neubauer reports financial support was provided by Schrankerl GmbH. Lukas Neubauer reports financial support was provided by Austrian Research Promotion Agency.

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Appendix A. Methodology

A.1. Dynamic time warping

The set of all allowed warping functions Φ depends on the type of time warping, which is usually parametrized by a step pattern. This pattern defines the allowed values of $\phi(k)$ given $\phi_k(1), \dots, \phi_k(k-1)$ for each k.

The basic properties of a warping function are as follows.

- **Monotonicity**. We have $\phi_{\mathbf{X}}(k) \le \phi_{\mathbf{X}}(k+1), \phi_{\mathbf{Y}}(k) \le \phi_{\mathbf{Y}}(k+1)$ for all k. This allows for only reasonable matchings where the order of the sequences is unchanged.
- **Slope**. The possible (local) slope of ϕ is given by
$$Q_{k_s} = \frac{\sum_{i=1}^{T_{k_s}} q_i^{(k_s)}}{\sum_{i=1}^{T_{k_s}} p_i^{(k_s)}},$$
where s denotes the step pattern for the warping function, T_{k_s} gives the number of allowed steps in the kth step pattern path, and $p_i^{(k_s)}, q_i^{(k_s)}$ gives the actual size of steps in the $\mathbf{X}$ and $\mathbf{Y}$ directions, respectively. Thus, we have that $p_i^{(k_s)} = \phi_{\mathbf{X}}(i+1) - \phi_{\mathbf{X}}(i)$ for any i, and $q_i^{(k_s)}$ analogously.
The slope is therefore constrained by $\min_{k_s} Q_{k_s} \le Q_{k_s} \le \max_{k_s} Q_{k_s}$.
- **Normalizability**. Depending on ϕ, the DTW distance can also be normalized by $M_\phi = \sum_{k=1}^{K} w_\phi(k)$. That is, we have $\text{nDTW}(\mathbf{X}, \mathbf{Y}) := \text{DTW}(\mathbf{X}, \mathbf{Y})/M_\phi$.

More properties arise when moving to more concrete warping functions.

A.1.1. Symmetric DTW

The most common type is the so-called symmetric matching pattern, where

- the endpoints must be matched, meaning we have $\phi(1) = (\phi_{\mathbf{X}}(1), \phi_{\mathbf{Y}}(1)) = (1,1)$ and $\phi(K) = (\phi_{\mathbf{X}}(K), \phi_{\mathbf{Y}}(K)) = (n,m)$;
- all points must be matched, i.e. $\phi_{\mathbf{X}}(k+1) - \phi_{\mathbf{X}}(k) \le 1$, $\phi_{\mathbf{Y}}(k+1) - \phi_{\mathbf{Y}}(k) \le 1$ (this can also be seen as a continuity constraint);
- allowed steps to get to (i,j) are $(i-1,j) \to (i,j)$, $(i,j-1) \to (i,j)$, $(i-1,j-1) \to (i,j)$ (see also Eq. (A.1)), implying that
- the slope of ϕ is unconstrained, allowing for an arbitrary amount of time stretching or compression because $Q = q/p$, $p,q \in \{0,1\}$ and $0 \le Q \le \infty$; and
- the weights are chosen to be $w_\phi(k) = \phi_{\mathbf{X}}(k) - \phi_{\mathbf{X}}(k-1) + \phi_{\mathbf{Y}}(k) - \phi_{\mathbf{Y}}(k-1)$ such that $M_\phi = n+m$.

In fact, the step pattern can be written using a dynamic programming (Bellman & Dreyfus, 1962) approach:

$$g(i,j) = \min \begin{pmatrix} g(i,j-1) + d(i,j) \\ g(i-1,j-1) + 2d(i,j) \\ g(i-1,j) + d(i,j) \end{pmatrix}, \quad 1 \le i \le n,\ 1 \le j \le m, \tag{A.1}$$

such that $g(1,1) = w(1)d(1,1) = 2d(1,1)$ and $g(n,m) =: \mathrm{DTW}_{\mathrm{sym}}(\mathbf{X},\mathbf{Y})$, where $n = |\mathbf{X}|, m = |\mathbf{Y}|$. The resulting matrix G of Eq. (A.1) is called the warping matrix. The corresponding warping path can be extracted from these recursive calculations by backtracking.

This distance is naturally symmetric and positive definite. However, it is not a metric, since the triangle inequality is usually not fulfilled.

This pattern type can be useful when considering sequences of similar lengths. However, for differently sized sequences, this might not work reasonably anymore, due to the above constraints.

A.1.2. Asymmetric DTW

We are particularly interested in subsequence matching where we relax the endpoint constraints. This is also denoted as open-begin open-end (OBE) matching (Giorgino, 2009). The corresponding optimization problem looks as follows:

$$\mathrm{DTW}_{\mathrm{OBE}}(\mathbf{X},\mathbf{Y}) := \min_{1\le p\le q\le m} \mathrm{DTW}(\mathbf{X},\mathbf{Y}^{(p,q)}),$$

where $\mathbf{Y}^{(p,q)} = (\mathbf{Y}_p,\dots,\mathbf{Y}_q)' \in \mathbb{R}^{(q-p+1)\times d}$ is a subsequence of $\mathbf{Y}$, and $n \le m$. This type of matching requires an asymmetric matching pattern. The most basic one is given by

$$g(i,j) = \min\begin{pmatrix} g(i-1,j)+d(i,j) \\ g(i-1,j-1)+d(i,j) \\ g(i-1,j-2)+d(i,j)\end{pmatrix}, \quad 1\le i\le n, p\le j\le q, \tag{A.2}$$

implying that continuity is not imposed anymore. In fact, information of $\mathbf{Y}$ can be skipped. The slope is constrained by $\min Q = 0$, $\max Q = 2$. Further, the distance is only normalizable by n, since we consider subsequences of $\mathbf{Y}$ of different lengths. Because of the subsequence matching, we have an initial value of $g(1,p) = d(1,p)$ and $g(n,q) =: \mathrm{DTW}_{\mathrm{asym,\ OBE}}(\mathbf{X},\mathbf{Y})$.

A.1.3. Other step patterns

Many more step patterns exist with different types of slope constraints and other properties. Unfortunately, it is often not very clear which step pattern really is the most appropriate one to use. A general family of step patterns is defined in Sakoe and Chiba (1978) by the notion of *symmetricPx*, and *asymmetricPx*, where x controls the slope parameter. The usual values are 0, 0.5, 1, and 2. The larger the slope can possibly be, the more complicated the dynamic programming equations tend to get. As a limiting step pattern, one obtains the *rigid* step pattern for $x \to \infty$. As suggested by Giorgino (2009), this step pattern is only reasonable when considering an OBE matching, since it finds the most appropriate subsequence without any gaps and, in fact, does not perform any time warping.

To summarize, DTW is a very flexible sequence matching method which yields the optimal matching, as well as an associated distance, between any two sequences.

A.2. DTW averaging

As with all dissimilarity measures, DTW can also be used to find a barycenter of a set of sequences $\mathcal{S}$. A barycenter is usually defined to minimize the sum of distances in the set. Generally, $c \in X$ for some metric space (X,δ) is called a barycenter of $Y \subset X$ if

$$\sum_{x\in Y}\delta(c,x) \le \sum_{x\in Y}\delta(y,x), \tag{A.3}$$

for any $y \in X$.

In terms of time series and DTW, we have $Y = \mathcal{T}$ as the finite set of time series of interest, $X \supset Y$ as the set of all possible time series with some maximum length, and $\delta = \mathrm{DTW}$. However, the space of possible solutions is very large, which makes the search for the average difficult. Therefore, approximative solutions have been developed.

In the context of DTW, it is not straightforward to provide a definition for an average. For the symmetric case of DTW, Gupta et al. (1996) considered pairwise, coordinate-wise averaging, where two sequences are averaged to one sequence until only one sequence is left. This method is easy to implement, but a big downside is that it heavily depends on the order of sequences. A different approach, introduced by Petitjean et al. (2011), is global averaging. Starting from an initial sequence, the average is updated in each iteration based on all matchings of all sequences in the set. As mentioned in their paper, this heuristic naturally reduces the sum of the warping distances to the average in Eq. (A.3).

One aspect to consider is the length of the averaging sequence. In pairwise averaging, this average can grow twice as long in each step. In global averaging, the length of the resulting average is fixed to the length of the initial sequence.

We adapt the global averaging methodology to the asymmetric DTW use case as follows. Denote by $\mathcal{T}$ a set of time series of different lengths. Our aim is to have a longer barycenter than the time series of interest. Hence, as the initial average time series we take the longest one available, i.e. $\mathbf{C}_0 = \mathrm{argmax}_{\mathbf{X}\in\mathcal{T}}|\mathbf{X}|$. Then, we compute $\mathbf{C}_i$, $i = 1,2,\dots$ iteratively:

- Compute $\mathrm{DTW}_{\mathrm{asym,\ OBE}}(\mathbf{X},\mathbf{C}_i)$ for all $\mathbf{X}\in\mathcal{T}$ to obtain $\phi^{(\mathbf{X})}(k) := (\phi_{\mathbf{X}}(k),\phi_{\mathbf{C}_i}(k))$ for $k = 1,\dots,K_\phi$.
- For each time step $t = 1,\dots,|\mathbf{C}_0|$ of $\mathbf{C}_i$, denoted by $\mathbf{C}_{i,t}$, let $\mathbb{C}_{i,t} := \left\{\mathbf{X}_j : \mathbf{X}\in\mathcal{T}, \phi^{(\mathbf{X})} = (j,t)\right\}$ be the set of all associated $\mathbf{X}$ time steps.
- Update each time step of the averaging sequence by

$$\mathbf{C}_{i+1,t} = \begin{cases} \frac{1}{|\mathbb{C}_{i,t}|}\sum_{\mathbf{Z}\in\mathbb{C}_{i,t}}\mathbf{Z} & \text{if } |\mathbb{C}_{i,t}|>0, \\ \mathbf{C}_{i,t} & \text{otherwise.}\end{cases}$$

We perform this iteration for a fixed number of I times or until the sum in Eq. (A.3) does not decrease anymore. We now denote the averaging time series as $\mathrm{aDBA}(\mathcal{T}) := \mathbf{C}_{\min(S,I)}$, where $S > S^*$ is the time of no further reduction after a start-up period S^*, and the averaging function as $\mathrm{aDBA} : \mathcal{T} \to \mathcal{Z}$, where $\mathcal{Z}$ denotes the set of all possible time series.

A.3. Theoretical motivation

We want to motivate our approach of using the DTW distance. For that we consider just a simple case. Let X

be an *ANN* model (Hyndman et al., 2008). An *ANN* model is also known as exponential smoothing. It does not have any trend or seasonality component, and for a time series $(X_t, t = 1, \dots, n)$ it is given by the recursion

$$l_t^X = \alpha X_t + (1-\alpha) l_{t-1}^X, \tag{A.4}$$
$$\hat{X}_{t+h|t} = l_t^X, \quad h > 0,$$

where l^X denotes the level component of the model, which is also equal to the flat forecast $\hat{X}_{t+h|t}$ for any $h > 0$. The model parameter α is usually found by minimizing the sum of squared forecast errors or by maximum likelihood.

A.3.1. Theoretical DTW computation

Let X, Y be two independent *ANN* models. Without loss of generalization, we assume that both initial values are equal to 0, i.e. $l_0^X = l_0^Y = 0$ (otherwise we could just look at $X - l_0^X$). We write $X = (X_1, \dots, X_n)' \sim ANN(\alpha_X, \sigma_X^2)$ and $Y = (Y_1, \dots, Y_n)' \sim ANN(\alpha_Y, \sigma_Y^2)$. For both, we consider an equal length of $n = |X| = |Y|$. Considering Eq. (A.4), we can rewrite the recursive equation using a state-space representation as

$$X_t = l_{t-1}^X + \epsilon_t, \qquad Y_t = l_{t-1}^Y + \eta_t$$
$$l_t^X = l_{t-1}^X + \alpha_X \epsilon_t, \quad l_t^Y = l_{t-1}^Y + \alpha_Y \eta_t,$$

where the innovations $\epsilon_t \overset{iid}{\sim} N(0, \sigma_X^2)$, $\eta_t \overset{iid}{\sim} N(0, \sigma_Y^2)$ are also assumed to be pairwise independent for $t = 1, \dots, n$. The innovations are the one-step-ahead forecast errors given by $\epsilon_t = X_t - l_{t-1}^X$. The states l^X are latent, and only X_t itself is observable.

Next, we give an explicit expression for the asymmetric DTW distance between X and Y when considering the squared L^2 cross-distance. We denote X to be independent of Y if and only if X_i is independent of Y_j for all i, j. This is a simple assumption, but the following results can be easily extended to a more general setup.

Lemma 1. *Let $X \sim ANN(\alpha_X, \sigma_X^2)$, $Y \sim ANN(\alpha_Y, \sigma_Y^2)$ be two independent and centered exponential smoothing processes of length n. Let $d(i,j) := \mathbb{E}[(X_i - Y_j)^2]$ denote the cross-distance between X and Y.*

Then the asymmetric DTW distance is given by

$$DTW_{asym}(X,Y) = \sigma_X^2 \left(n + \binom{n}{2} \alpha_X^2 \right) + \sigma_Y^2 \left(n + \left\lfloor \frac{n^2}{4} \right\rfloor \alpha_Y^2 \right). \tag{A.5}$$

Proof. First, note that the cross-distance is given by

$$\begin{aligned} d(i,j) &= \mathbb{E}[X_i^2] + \mathbb{E}[Y_j^2] \\ &= \sigma_X^2(1 + (i-1)\alpha_X^2) + \sigma_Y^2(1 + (j-1)\alpha_Y^2), \end{aligned}$$

due to the independence of X_i, Y_j for every i, j. Next, we need to recursively compute the warping matrix G as in Eq. (A.2). We have $g(1,1) = d(1,1) = \sigma_X^2 + \sigma_Y^2$. Due to the cross-distance increasing in both i and j, we can easily solve Eq. (A.2) by

$$\begin{aligned} g(i,j) &= \min \begin{pmatrix} g(i-1,j) \\ g(i-1,j-1) \\ g(i-1,j-2) \end{pmatrix} + d(i,j) \\ &= g\left(\tilde{i}, \tilde{j}\right) + \sum_{k=0}^{i-\tilde{i}-1} d(i-k, j-2k), \end{aligned}$$

where $\tilde{i}$, $\tilde{j}$ denotes the indices where the recursion needs more specific computation. We need this distinction because for small i, j, the minimum value is different than expressed in the sum, because some of the values are not assigned. In detail, we have that

$$g\left(\tilde{i}, \tilde{j}\right) = \begin{cases} \sum_{k=0}^{\tilde{i}-1} d(\tilde{i}-k, 1) & \text{if } \tilde{i} > 0, \tilde{j} = 1, \\ g(\tilde{i}-1, 1) + d(\tilde{i}, 2) & \text{if } \tilde{i} > 1, \tilde{j} = 2, \\ \text{NA} & \text{if } \tilde{i} = 1, \tilde{j} > 1. \end{cases}$$

Altogether, we obtain, by induction,

$$g(i,j) = \begin{cases} i(\sigma_X^2 + \sigma_Y^2) + \binom{i}{2}\sigma_X^2\alpha_X^2 + \lfloor \frac{i^2}{4} \rfloor \sigma_Y^2 \alpha_Y^2 & \text{if } 2i - j \geq 1, \\ \text{NA} & \text{otherwise.} \end{cases} \tag{A.6}$$

Setting $i = j = n$ finishes the proof. □

A.3.2. Relation to Wasserstein distance

Since we are interested in the one-step-ahead forecast, we may look at the corresponding forecast distributions. The forecast distributions are given by the conditional distributions of $\hat{X}_{n+1|n} = X_{n+1}|l_n^X \sim N(l_n^X, \sigma_X^2)$ and $\hat{Y}_{n+1|n} = Y_{n+1}|l_n^Y \sim N(l_n^Y, \sigma_Y^2)$, respectively. Now we want to measure the distance between those two distributions. For that we use the 2-Wasserstein distance, which is defined as follows (Villani, 2009). Let μ, ν be two measures and $\pi(\mu, \nu)$ be the set of all couplings of μ and ν. Then

$$W_2(\mu, \nu) = \sqrt{\inf_{\gamma \in \pi(\mu,\nu)} \int \|x - y\|^2 d\gamma(x,y)}.$$

In the case of random variables we can also write

$$W_2(X,Y) = \sqrt{\inf\{\mathbb{E}[\|\tilde{X} - \tilde{Y}\|^2] : (\tilde{X}, \tilde{Y}) \in \pi(X,Y)\}},$$

for any two random variables X, Y. If both are Gaussian, i.e. $X \sim N(\mu_1, \Sigma_1)$ and $Y \sim N(\mu_2, \Sigma_2)$, then the squared 2-Wasserstein distance is readily computed by

$$\begin{aligned} W_2^2(X,Y) &= \|m_1 - m_2\|^2 \\ &\quad + \operatorname{tr}\left(\Sigma_1 + \Sigma_2 - 2(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2})^{1/2}\right). \end{aligned} \tag{A.7}$$

Details are available in the work of Olkin and Pukelsheim (1982). Applying Eq. (A.7) to the forecast distributions of the *ANN* models, we can calculate the 2-Wasserstein distance between $\hat{X}_{n+1|n}$ and $\hat{Y}_{n+1|n}$, yielding

$$W_2^2(\hat{X}_{n+1|n}, \hat{Y}_{n+1|n}) = (l_n^X - l_n^Y)^2 + (\sigma_X - \sigma_Y)^2 \tag{A.8}$$

Thus, both Eqs. (A.5) and (A.8) are quadratic in the model parameters of *ANN*, allowing us to give following theorem.

Theorem 2. *Let $\mathcal{X}$ be the space of independent ANN processes of length $n > n(p)$ for some $p \in (0,1)$, equipped with the asymmetric DTW distance, and let $\mathcal{Y}$ be the space of*

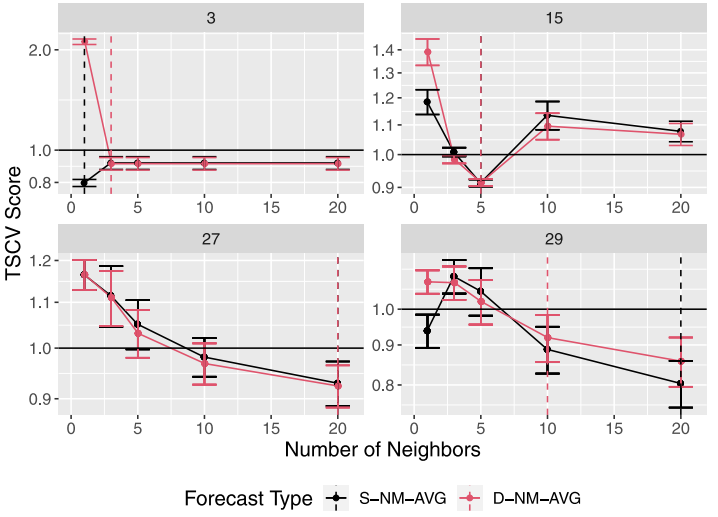


Fig. B.12. TSCV scores for simple no-model averaging. The dashed vertical lines indicate the optimal number of neighbors.

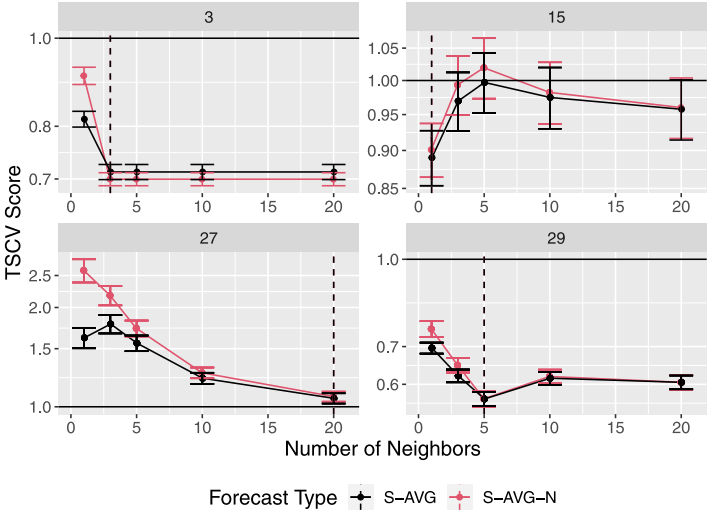


Fig. B.13. TSCV scores for simple averaging. The dashed vertical lines indicate the optimal number of neighbors.

corresponding Gaussian forecast distributions equipped with the squared 2-Wasserstein distance.

Then the map $\mathcal{X} \to \mathcal{Y} : X \mapsto \hat{X}_{n+1|n}$ *is Lipschitz-continuous with Lipschitz constant* $L < 1$ *and probability of at least p.*

Proof. Let $X, Y \in \mathcal{X}$ be two arbitrary *ANN* processes. We have by Eq. (A.8) that $W_2^2(\hat{X}_{n+1|n}, \hat{Y}_{n+1|n}) = (l_n^X - l_n^Y)^2 + (\sigma_X - \sigma_Y)^2$ for fixed states l_n^X, l_n^Y. Since l_n^X, l_n^Y are both realizations of independent Gaussian random variables, we obtain

$$\mathbb{P}\left((l_n^X - l_n^Y)^2 \leq q_p n(\alpha_X^2\sigma_X^2 + \alpha_Y^2\sigma_Y^2)\right) = p, \tag{A.9}$$

where q_p is the p-quantile of a $\chi^2(1)$ distribution. Then, using Eq. (A.9) yields

$$p = \mathbb{P}\left(\frac{W_2^2(\hat{X}_{n+1|n}, \hat{Y}_{n+1|n})}{\mathrm{DTW}_{\mathrm{asym}}(X, Y)} \leq \frac{q_p n(\sigma_X^2\alpha_X^2 + \sigma_Y^2\alpha_Y^2) + (\sigma_X - \sigma_Y)^2}{\sigma_X^2\left(n + \binom{n}{2}\alpha_X^2\right) + \sigma_X^2\left(n + \lfloor\frac{n^2}{4}\rfloor\alpha_Y^2\right)}\right) \leq$$

$$\mathbb{P}\left(\frac{W_2^2(\hat{X}_{n+1|n}, \hat{Y}_{n+1|n})}{\mathrm{DTW}_{\mathrm{asym}}(X, Y)} \leq\right.$$

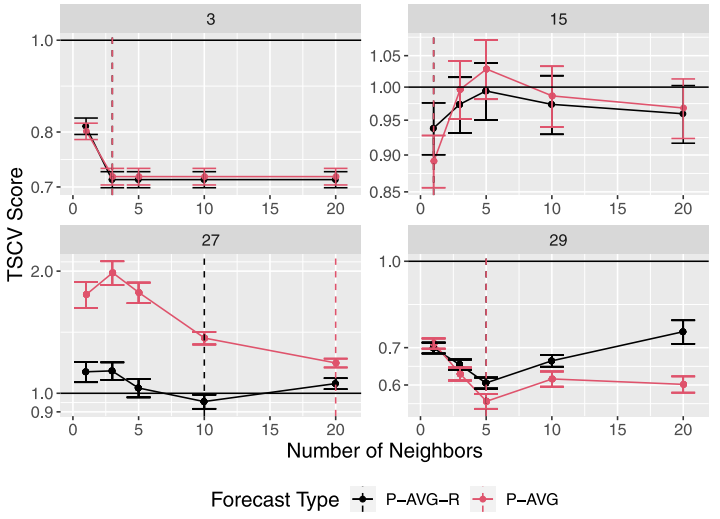


Fig. B.14. TSCV scores for performance-based averaging. The dashed vertical lines indicate the optimal number of neighbors.

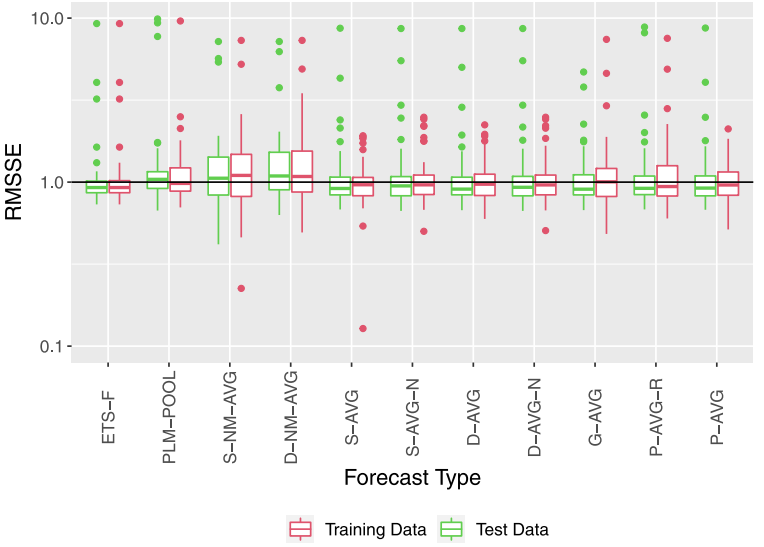


Fig. C.15. Global RMSSE with respect to random-walk forecasts.

$$\left.\frac{q_p(\sigma_X^2(1+n\alpha_X^2)+\sigma_Y^2(1+n\alpha_Y^2))}{\sigma_X^2\left(n+\binom{n}{2}\alpha_X^2\right)+\sigma_X^2\left(n+\lfloor\frac{n^2}{4}\rfloor\alpha_Y^2\right)}\right) \leq$$

$$\mathbb{P}\left(\frac{W_2^2(\hat{X}_{n+1|n},\hat{Y}_{n+1|n})}{\mathrm{DTW}_{\mathrm{asym}}(X,Y)}<1\right)$$

using that $\binom{n}{2}, \lfloor n^2/4\rfloor > n$ for $n > n(p)$. Thus, the map $X \mapsto \hat{X}_{n+1|n}$ is Lipschitz-continuous with constant $L < 1$ and probability at least p. □

Remark. If we want to have Lipschitz continuity with at least 95% probability, then $q_{0.95} < 4$ and Theorem 2 holds with $n > 16$.

Remark. This result also holds when looking at the normalized DTW measure with Lipschitz constant $L > 1$ and is therefore not a contraction anymore.

Remark. This also tells us that close time series in terms of DTW are also close in their corresponding forecast distribution both in mean and variance. Further, it assures us that small changes in the time series only affect the difference in forecast distributions by a small amount.

More detailed results are obtained when considering the mean forecasts given by $\mathbb{E}[X_{n+1|n}] := \mathbb{E}[X_{n+1}|I_n^X] = I_n^X \sim N(0, n\sigma_X^2\alpha_X^2)$ and $\mathbb{E}[Y_{n+1|n}] = I_n^Y \sim N(0, n\sigma_Y^2\alpha_Y^2)$. The

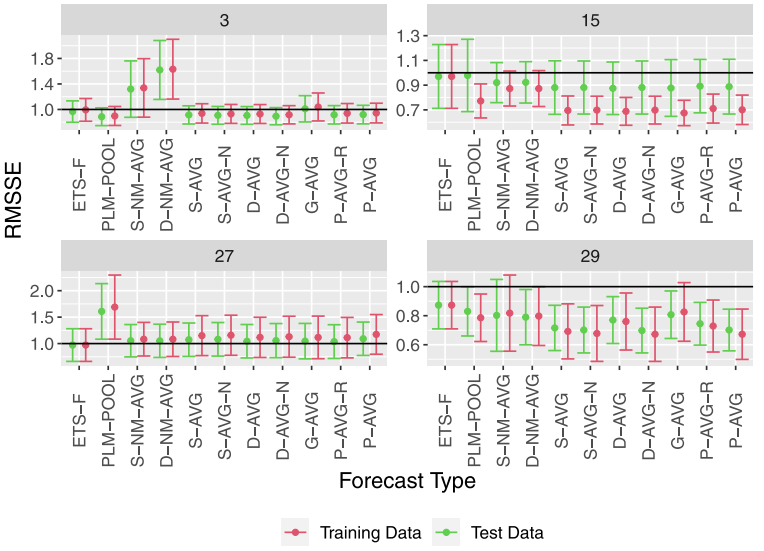


Fig. C.16. Individual RMSSE with respect to random-walk forecasts.

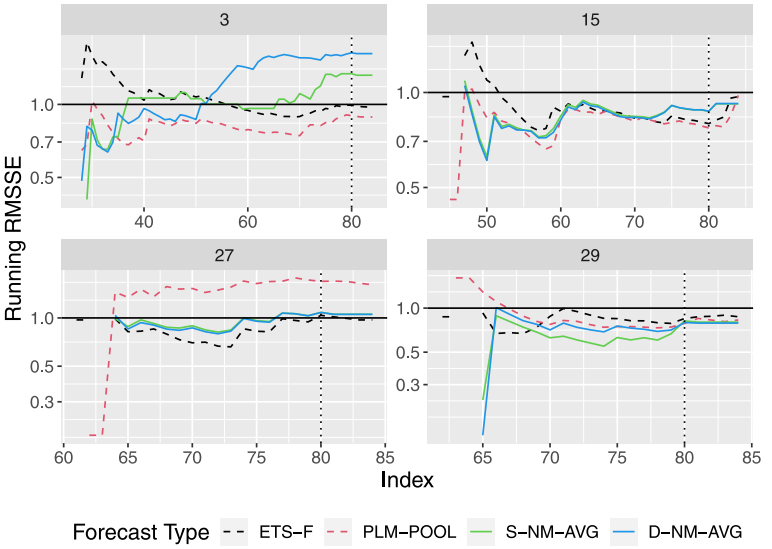


Fig. C.17. Running RMSSE with respect to random-walk forecasts for simple no-model averaging.

corresponding 2-Wasserstein distance is computed to be

$$\begin{aligned} W_2^2(\mathbb{E}[X_{n+1|n}], \mathbb{E}[Y_{n+1|n}]) &= n\sigma_X^2\alpha_X^2 + n\sigma_Y^2\alpha_Y^2 \\ &\quad - 2\sqrt{n^2\sigma_X^2\sigma_Y^2\alpha_X^2\alpha_Y^2} \\ &= n(\sigma_Y\alpha_X - \sigma_Y\alpha_Y)^2. \end{aligned}$$

Theorem 3. *Let $\mathcal{X}$ be the space of independent ANN processes of length $n > 5$ equipped with the asymmetric DTW distance, and let $\mathcal{Y}$ be the space of corresponding Gaussian forecast distributions equipped with the squared 2-Wasserstein distance.*

Then the map $\mathcal{X} \to \mathcal{Y} : X \mapsto \mathbb{E}[X_{n+1|n}]$ is Lipschitz-continuous with Lipschitz constant $L < 1$.

Proof. Let $X, Y \in \mathcal{X}$ be two arbitrary *ANN* processes. We have that

$$\begin{aligned} &\frac{W_2^2(\mathbb{E}[X_{n+1|n}], \mathbb{E}[Y_{n+1|n}])}{\mathrm{DTW}_{\mathrm{asym}}(X, Y)} \\ &= \frac{n(\sigma_X\alpha_X - \sigma_Y\alpha_Y)^2}{\sigma_X^2\left(n + \binom{n}{2}\alpha_X^2\right) + \sigma_X^2\left(n + \lfloor\frac{n^2}{4}\rfloor\alpha_Y^2\right)} \\ &< 1, \end{aligned}$$

using that $\binom{n}{2}, \lfloor n^2/4 \rfloor > n$ for $n > 5$. Thus, the map $X \mapsto \mathbb{E}[X_{n+1|n}]$ is Lipschitz-continuous with constant $L < 1$. □

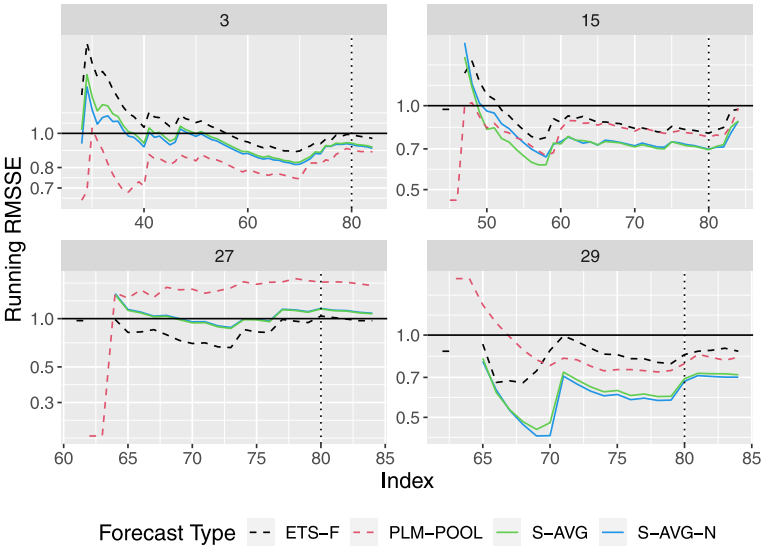


Fig. C.18. Running RMSSE with respect to random-walk forecasts for simple averaging.

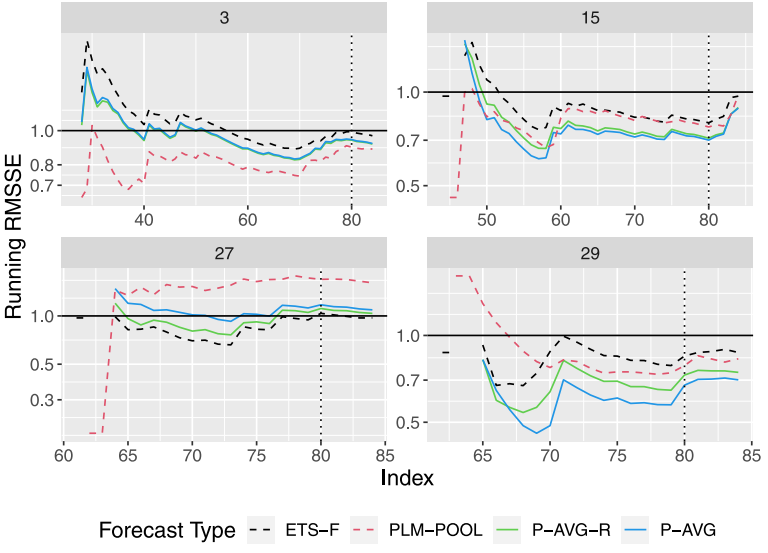


Fig. C.19. Running RMSSE with respect to random-walk forecasts for performance-based averaging.

A.3.3. Reduction of mean squared error

Another result is about the relation of DTW and the mean squared error of a convex combination of the mean forecasts. Let $Z_{n+1|n}(w) := w\mathbb{E}[X_{n+1|n}] + (1-w)\mathbb{E}[Y_{n+1|n}]$ for $w \in [0,1]$. We have $Z_{n+1|n}(w) = wl_n^X + (1-w)l_n^Y$. However, in practice, the states l_n are not known and need to be estimated by actually estimating the smoothing parameter α. To this end, assume there exists unbiased estimators $\hat{\alpha}_X$, $\hat{\alpha}_Y$ for α_X, α_Y, respectively. We further assume they have finite second moment. The corresponding estimating forecasts are given by $\hat{z}(w)$. Then, under certain conditions for the estimation errors made for α_X, α_Y, we have following result.

Theorem 4. *Let $X \sim ANN(\alpha_X, \sigma_X^2)$, $Y \sim ANN(\alpha_Y, \sigma_Y^2)$ be two independent and centered exponential smoothing processes of length n and known variances σ_X^2, σ_Y^2. Then a convex combination of the forecasts $\hat{Z}$ reduces the mean squared error:*

$$\mathbb{E}[(X_{n+1} - \hat{z}(w))^2] \leq \mathbb{E}[(X_{n+1} - \hat{l}_n^X)^2],$$

if $MSE(\hat{\alpha}_Y) \leq \frac{\sigma_X^2}{2\sigma_Y^2}\big((1-w^2)MSE(\hat{\alpha}_X) - nDTW_{asym}(X,Y)/\sigma_X^2\big)$.

Proof. We have that

$$\begin{aligned}\mathbb{E}[(X_{n+1} - \hat{z}(w))^2] &= \mathbb{E}[(l_n^X + \epsilon_{n+1} - (w\hat{l}_n^X + (1-w)\hat{l}_n^Y))^2] \\ &= \sigma_X^2 + \mathbb{E}[(l_n^X - (w\hat{l}_n^X + (1-w)\hat{l}_n^Y))^2],\end{aligned}$$

using the independence of the error term ϵ_{n+1}. Further, we obtain

$$\begin{aligned}\mathbb{E}[(X_{n+1}-\hat{z}(w))^2] &\leq \mathbb{E}[(l_n^X - l_n^Y)^2] + w^2\mathbb{E}[(\hat{l}_n^X - \hat{l}_n^Y)^2] \\ &\quad + \mathbb{E}[(l_n^Y - \hat{l}_n^Y)^2] \\ &= n(\sigma_X^2\alpha_X^2 + \sigma_Y^2\alpha_Y^2) + w^2(\mathbb{E}[(\hat{l}_n^X)^2] \\ &\quad + \mathbb{E}[(\hat{l}_n^Y)^2]) + \mathbb{E}[(l_n^Y - \hat{l}_n^Y)^2].\end{aligned}$$

We can quickly compute the last terms above by

$$\begin{aligned}\mathbb{E}[(l_n^Y - \hat{l}_n^Y)^2] &= \int \mathbb{E}[(l_n^Y - \hat{l}_n^Y)^2|\hat{\alpha}_Y = a]\mathbb{P}(\hat{\alpha}_Y = a)da \\ &= n\sigma_Y^2 \int (a-\alpha_Y)^2\mathbb{P}(\hat{\alpha}_Y = a)da \\ &= n\sigma_Y^2\mathrm{MSE}(\hat{\alpha}_Y), \quad \text{and} \\ \mathbb{E}[(\hat{l}_n^X)^2] &= n\sigma_X^2\mathbb{E}[\hat{\alpha}_X^2] \\ &= n\sigma_X^2(\mathrm{MSE}(\hat{\alpha}_X) + \alpha_X^2).\end{aligned}$$

In total we get that

$$\begin{aligned}\mathbb{E}[(X_{n+1}-\hat{z}(w))^2] &\leq n\sigma_X^2\big((1+w^2)\alpha_X^2 + w^2\mathrm{MSE}(\hat{\alpha}_X)\big) + \\ &\quad n\sigma_Y^2\big((1+w^2)\alpha_Y^2 + 2\mathrm{MSE}(\hat{\alpha}_Y)\big) \\ &\overset{!}{\leq} \mathbb{E}[(X_{n+1} - \hat{l}_n^X)^2] \\ &= n\sigma_X^2\mathrm{MSE}(\hat{\alpha}_X).\end{aligned}$$

Using that $n\sigma_X^2(1+w^2)\alpha_X^2 + n\sigma_Y^2(1+w^2)\alpha_Y^2 \leq \mathrm{DTW}_{\mathrm{asym}}(X,Y)$, this finally yields

$$\begin{aligned}\mathbb{E}[(X_{n+1}-\hat{z}(w))^2] &\leq \mathrm{DTW}_{\mathrm{asym}}(X,Y) + n(w^2\sigma_X^2\mathrm{MSE}(\hat{\alpha}_X) \\ &\quad + 2\sigma_Y^2\mathrm{MSE}(\hat{\alpha}_Y)) \\ &\leq n\sigma_X^2\mathrm{MSE}(\hat{\alpha}_X),\end{aligned}$$

if $\mathrm{MSE}(\hat{\alpha}_Y) \leq \frac{\sigma_X^2}{2\sigma_Y^2}\left((1-w^2)\mathrm{MSE}(\hat{\alpha}_X) - \mathrm{nDTW}_{\mathrm{asym}}(X,Y)/\sigma_X^2\right)$. □

Remark. In practice, the previous theorems tell us that if X, Y are close in terms of DTW and, additionally, the estimation error made when estimating α_Y is smaller than the error made for α_X, then the convex combination forecast improves the point forecast for X_{n+1}.

The condition of $\mathrm{MSE}(\hat{\alpha}_Y) \ll \mathrm{MSE}(\hat{\alpha}_X)$ might have various reasons. In an application, the fit of the model might be better for Y than for X, resulting in better estimation of α.

A.3.4. Conclusions

These theoretical results give us arguments of our methodology for the most simple cases of models. However, the arguments might be extended to a broader family of models, as given in Hyndman et al. (2008). In practice, we also need to use open-begin open-end matching, since the asymmetric DTW measure cannot be computed once the reference time series is too long. Still, similar arguments should hold and motivate our approach to using DTW neighborhoods and performing model averaging. The theory does not give us hints how to choose the optimal weights, so we propose the weights of Section 2.3.

Appendix B. Further TSCV evaluation plots

See Figs. B.12–B.14

Appendix C. Further evaluation plots

See Figs. C.15–C.19

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